

Human review packet: MSVV07AdWords

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: MSVV07AdWords
Paper id: MSVV07AdWords
Source version: Journal of the ACM, 2007
Source record: audit/paper_statement_map.json
Rows: 76
Generated: 2026-08-18
Source URL: [official source](#)

Review status: 0/76 source-claim screens; 0/335 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper MSVV07AdWords --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

Contents

- [Semantic library prerequisites \(335; not paper claims\)](#)
 - [EconCSLib.Online.AdWordsInstance](#)
 - [EconCSLib.Online.AdWordsInstance.Assignment](#)
 - [EconCSLib.Online.AdWordsInstance.bid](#)
 - [EconCSLib.Online.AdWordsInstance.budget](#)
 - [EconCSLib.Online.AdWordsInstance.spend](#)
 - [EconCSLib.Online.AdWordsInstance.CanAssign](#)
 - [EconCSLib.Online.AdWordsInstance.ChoiceRule](#)
 - [EconCSLib.Online.AdWordsInstance.Feasible](#)
 - [EconCSLib.Online.AdWordsInstance.HistoryState](#)
 - [EconCSLib.Online.AdWordsInstance.HistoryState.assignment](#)
 - [EconCSLib.Online.AdWordsInstance.HistoryState.mk](#)
 - [EconCSLib.Online.AdWordsInstance.HistoryState.seen](#)
 - [EconCSLib.Online.AdWordsInstance.balanceDiscount](#)
 - [EconCSLib.Online.AdWordsInstance.spentFraction](#)
 - [EconCSLib.Online.AdWordsInstance.balanceScore](#)
 - [EconCSLib.Online.AdWordsInstance.IsBalanceChoice](#)
 - [EconCSLib.Online.AdWordsInstance.revenue](#)
 - [EconCSLib.Online.AdWordsInstance.IsOptimalAssignment](#)
 - [EconCSLib.Online.AdWordsInstance.NonnegativeBids](#)
 - [EconCSLib.Online.AdWordsInstance.NonnegativeBudgets](#)
 - [EconCSLib.Online.AdWordsInstance.PositiveBudgets](#)
 - [EconCSLib.Online.AdWordsInstance.assignQuery](#)
 - [EconCSLib.Online.AdWordsInstance.exists_balanceChoice_of_exists_canAssign](#)
 - [EconCSLib.Online.AdWordsInstance.balanceChoiceRule](#)
 - [EconCSLib.Online.AdWordsInstance.emptyAssignment](#)
 - [EconCSLib.Online.AdWordsInstance.exists_optimalAssignment](#)
 - [EconCSLib.Online.AdWordsInstance.historyFinset](#)

- EconCSLib.Online.AdWordsInstance.initialHistoryState
- EconCSLib.Online.AdWordsInstance.maxBidForQuery
- EconCSLib.Online.AdWordsInstance.mk
- EconCSLib.Online.AdWordsInstance.offlineOptimumAssignment
- EconCSLib.Online.AdWordsInstance.offlineOptimumValue
- EconCSLib.Online.AdWordsInstance.stepHistoryState
- EconCSLib.Online.AdWordsInstance.runHistoryStateFrom
- EconCSLib.Online.AdWordsInstance.runHistoryState
- EconCSLib.Online.AdWordsInstance.runAssignment
- EconCSLib.Online.AdWordsInstance.withEffectiveBids
- EconCSLib.Online.AdWordsInstance.withAdvertiserWeights
- EconCSLib.Online.AdWordsInstance.withAvailability
- EconCSLib.Online.AdWordsInstance.withClickThroughRates
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate
- EconCSLib.Online.theorem9BidderSpendUpperBound
- EconCSLib.Online.theorem9NormalizedRevenueUpperBound
- EconCSLib.Online.theorem9EligibleBidders
- EconCSLib.Online.theorem9ActualEligibleBidders
- EconCSLib.Online.theorem9ObservedPrefix
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.normalizedRevenue
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.prefixAllocation
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.normalizedRevenue_le_cappedPrefixAllocationSpend
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.revenueBound_le_ratio
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.toObservedPrefixAllocationRevenueCertificate
- EconCSLib.Online.BMatchingIntegralPrefixChoice
- EconCSLib.Online.BMatchingIntegralPrefixChoice.Feasible
- EconCSLib.Online.BMatchingIntegralPrefixAlgorithm
- EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation
- EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_sum_le_one
- EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_zero_of_not_visible
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.allocation
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.allocation_zero_of_ineligible
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.normalizedRevenue
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.revenueBound_le_ratio
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.round_allocation_sum_le_one
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.allocation
- EconCSLib.uniformPMF
- EconCSLib.Online.uniformPermutationDistribution
- EconCSLib.pmfExp
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.expectedAllocation_le
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.normalizedRevenue
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.revenueBound_le_ratio
- EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate
- EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.allocation
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.allocation_zero_of_ineligible
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue

- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.revenueBound_le_ratio
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.round_allocation_sum_le_one
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.mk
- EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetricPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate.normalizedRevenue
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.allocation
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.revenueBound_le_ratio
- EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate
- EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily
- EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.mk
- EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.prefixAllocation
- EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.normalizedRevenue
- EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily
- EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.choice
- EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.mk
- EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.toFeasiblePrefixRuleFamily
- EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.normalizedRevenue
- EconCSLib.Online.MSVV07Appendix.printedIndividualBudget
- EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible
- EconCSLib.Online.MSVV07Appendix.derivedKappaU
- EconCSLib.Online.MSVV07Appendix.derivedKappaY
- EconCSLib.Online.MSVV07Appendix.derivedKappaZ
- EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator
- EconCSLib.Online.MSVV07Appendix.derivedKappaB
- EconCSLib.Online.MSVV07Appendix.derivedKappaT
- EconCSLib.Online.MSVV07Appendix.derivedKappaX
- EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue
- EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue
- EconCSLib.Online.MSVV07Appendix.derivedKappaRatio
- EconCSLib.Online.MSVV07Appendix.instDecidable_mSVV07AdWords
- EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue
- EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail
- EconCSLib.Online.MSVV07Appendix.printedPhaseThreeServedMass
- EconCSLib.Online.MSVV07Appendix.printedContinuousOnlineValue
- EconCSLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate
- EconCSLib.Online.MSVV07Appendix.printedPhaseOneEligible
- EconCSLib.Online.MSVV07Appendix.printedPhaseOneRawBid
- EconCSLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore
- EconCSLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate
- EconCSLib.Online.MSVV07Appendix.printedPhaseThreeRawBid
- EconCSLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity
- EconCSLib.Online.MSVV07Appendix.printedPhaseTwoRawBid
- EconCSLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore
- EconCSLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore
- EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment
- EconCSLib.Online.MSVV07PaperFacing.PaperAssignment
- EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance
- EconCSLib.Online.MSVV07PaperFacing.PaperInstance
- EconCSLib.Online.MSVV07PaperFacing.PaperInterface.instDecidable_mSVV07AdWords
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff
- EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff
- EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio
- EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds

- EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperTradeoff
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperBalanceScore
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperCanAssign
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperFeasible
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperIsBalanceChoice
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperRevenue
- EconCSLib.Online.MSVV07PaperFacing.Proof.paperSmallBids
- EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive
- EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all
- EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9IntegralPrefixAlgorithm
- EconCSLib.Online.instDecidableEqBMatchingIntegralPrefixChoice
- EconCSLib.Online.instDecidableEqBMatchingIntegralPrefixAlgorithm
- EconCSLib.Online.instDecidablePredBMatchingIntegralPrefixChoiceFeasible
- EconCSLib.Online.instFintypeBMatchingIntegralPrefixChoice
- EconCSLib.Online.instFintypeBMatchingIntegralPrefixAlgorithm
- EconCSLib.Online.paper_adwords_theorem9_integral_prefix_algorithm_family
- EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9CappedNormalizedRevenue
- EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution
- EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9RandomizedOnlineAlgorithm
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.activeMassBeforeSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.allocationPerActiveBidder
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.fullMass
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteCandidateValue
- EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteCoeff
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.queryMassAtSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.revenue
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.spentBeforeSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.typeMass
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.candidateTests
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.scoreComparisons
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.winner
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution.finalSpend
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsAlphaCompetitive
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment.__f
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.query
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.spentBefore

- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.winner
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.candidateTests
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.decisionsRev
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.revenue
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.scoreComparisons
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.spent
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.mk
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistribution
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.regimeAt
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistributionAt
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceStep
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightQueryMassAtSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_active_mass_drop_realizes_type
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_budget_feasible
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_revenue_eq_N_sub_lpValue
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_slab_work_conservation
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_state_update
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_type_mass_partition
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate_row_tight
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidExecution
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan._f
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_nonneg
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_zero_of_ineligible
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_budget_feasible
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_eligible_equal_least_spend
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_finalSpend_eq_sum_allocation
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_allocation_conservation
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_full_or_all_eligible_saturated
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_spentBefore_eq_prior_allocations
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_state_update
- EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState

- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runBalanceOccurrences](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom.__f](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runReplicatedRankingProposal](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runWeightedBidProposal](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5Delta](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5OptQueryRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5OptTypeFiberRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QueryType](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6CurrentType](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardInstance](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardOfflineAssignment](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate](#)
- [EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9RoundBid](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperBalanceScore](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperCanAssign](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperFeasible](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperIsBalanceChoice](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperSmallBids](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperSpend](#)
- [EconCSLib.Online.MSVV07PaperFacing.paperTradeoff](#)
- [EconCSLib.Online.MSVV07PaperFacing.section6__next_highest_bid_alive](#)
- [EconCSLib.Online.MSVV07PaperFacing.section6__next_highest_bid_all](#)
- [EconCSLib.Online.MSVV07PaperFacing.theorem9IntegralPrefixAlgorithm](#)
- [EconCSLib.Online.MSVV07PaperFacing.theorem9CappedNormalizedRevenue](#)
- [EconCSLib.Online.MSVV07PaperFacing.theorem9HardDistribution](#)
- [EconCSLib.Online.MSVV07PaperFacing.theorem9RandomizedOnlineAlgorithm](#)
- [EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff](#)
- [EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff](#)
- [EconCSLib.Optimization.StandardMaxLP](#)
- [EconCSLib.Optimization.StandardMaxLP.mk](#)

- EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDelta
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDeltaCoeff
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteSuffix
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualRow
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible
- EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualObjective
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual
- EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate
- EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP
- EconCSLib.Optimization.IsMaximizerOn
- EconCSLib.Optimization.IsMinimizerOn
- EconCSLib.Optimization.StandardMaxLP.A
- EconCSLib.Optimization.StandardMaxLP.c
- EconCSLib.Optimization.StandardMaxLP.DualFeasible
- EconCSLib.Optimization.StandardMaxLP.b
- EconCSLib.Optimization.StandardMaxLP.PrimalFeasible
- EconCSLib.Optimization.StandardMaxLP.dualObjective
- EconCSLib.Optimization.StandardMaxLP.primalObjective
- EconCSLib.Sequence.SeqTendsTo
- Source claims (76)
 - paperFeasible_iffSpec
 - section2_competitive_ratio_definitionSpec
 - section3_discrete_tradeoff_formulaSpec
 - section3_discrete_tradeoff_monotonicitySpec
 - paperBalanceScore_formulaSpec
 - paperIsBalanceChoice_iffSpec
 - section3_greedy_tight_half_exampleSpec
 - section3_slab_and_current_slab_definitionSpec
 - section3_discrete_tradeoff_convergenceSpec
 - section3_discrete_balance_algorithmSpec
 - section3_equal_bid_discrete_msvv_is_balanceSpec
 - section6_current_type_definitionSpec
 - section6_generalized_current_type_balance_algorithmSpec
 - section6_beta_ij_definitionSpec
 - section6_beta_i_aggregate_definitionSpec
 - section6_alpha_i_definitionSpec
 - section6_alpha_total_definitionSpec
 - section6_beta_lower_bound_formulaSpec
 - section6_weighted_alpha_beta_inequalitySpec
 - section6_different_budgets_and_nonexhaustive_optimum_theorem8_finite_explicit_errorSpec
 - section6_next_highest_bid_all_formulaSpec
 - section6_next_highest_bid_alive_formulaSpec
 - section6_next_highest_alive_keep_alive_reductionSpec
 - section6_next_highest_bid_all_theorem8_finite_explicit_errorSpec
 - section6_next_highest_bid_alive_theorem8_finite_explicit_errorSpec
 - section6_click_through_rates_effective_bid_formulaSpec
 - section6_click_through_rates_theorem8_finite_explicit_errorSpec
 - section6_availability_theorem8_finite_explicit_errorSpec
 - section8_switching_query_distribution_modelSpec
 - section8_weighted_effective_bid_formulaSpec
 - section8_weighted_bid_algorithm_proposalSpec
 - section8_online_weight_adjustment_heuristicSpec

- `section8_replicated_ranking_algorithm_proposalSpec`
- `appendix_naive_three_phase_constructionSpec`
- `appendix_naive_algorithm_revenue_formulaSpec`
- `appendix_optimum_revenue_formulaSpec`
- `appendix_naive_strict_ratio_conclusionSpec`
- `appendix_kappa_counterexample_familySpec`
- `appendix_kappa_ratio_limit_infinitySpec`
- `appendix_kappa_ratio_limit_oneSpec`
- `theorem9HardDistribution_uniformSpec`
- `theorem9_hard_instance_round_bids_and_optimumSpec`
- `theorem9_base_instance_balance_revenue_claimSpec`
- `theorem9_expected_round_allocation_boundSpec`
- `theorem9_harmonic_spend_and_revenue_boundSpec`
- `theorem9CappedNormalizedRevenue_formulaSpec`
- `theorem9_no_randomized_online_algorithm_beats_msvv_ratioSpec`
- `section4_bidder_type_definitionSpec`
- `section4_type_count_x_definitionSpec`
- `section4_slab_spend_beta_formulaSpec`
- `section4_discretization_error_boundSpec`
- `section4_factor_revealing_primal_dual_lp_formulasSpec`
- `section4_lemma3_displayed_primal_dual_witnesses_optimalSpec`
- `section4_lemma1_balance_pays_no_later_slabSpec`
- `section4_lemma2_factor_revealing_lp_constraintSpec`
- `section4_lemma3_factor_revealing_lp_optimalitySpec`
- `section4_balance_exact_tightness_instanceSpec`
- `section5_tradeoff_revealing_primal_dual_lp_formulasSpec`
- `section5_bidder_type_definitionSpec`
- `section5_beta_slab_spend_definitionSpec`
- `section5_delta_prefix_definitionSpec`
- `section5_alg_query_revenue_definitionSpec`
- `section5_opt_query_revenue_definitionSpec`
- `section5_query_type_definitionSpec`
- `section5_query_slab_definitionSpec`
- `section5_type_fiber_alpha_accounting_formulaSpec`
- `section5_slab_fiber_beta_accounting_formulaSpec`
- `theorem8_dual_induced_tradeoff_formulaSpec`
- `theorem8_delta_dual_suffix_identitySpec`
- `theorem8_simple_proof_beta_recurrence_formulaSpec`
- `theorem8_simple_proof_weighted_alpha_beta_inequalitySpec`
- `theorem8_unspent_budget_bound_formulaSpec`
- `section5_lemma4_dual_optimal_from_primal_dual_matchSpec`
- `section5_lemma5_tradeoff_rhs_eq_base_add_deltaSpec`
- `section5_lemma6_per_query_tradeoffSpec`
- `section5_lemma7_weighted_perturbation_boundSpec`

Material library prerequisites

EconCSLib.Online.AdWordsInstance

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_1 → Type u_2 → Type (max u_1 u_2)

Exact Lean library declaration

```
structure AdWordsInstance (Advertiser Query : Type*) where
  budget : Advertiser → ℝ
  bid : Advertiser → Query → ℝ
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.Assignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_3) → (Query : Type u_4) → Query → Option Advertiser

Exact Lean library declaration

```
abbrev Assignment (Advertiser Query : Type*) :=  
  Query → Option Advertiser
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.bid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (self : EconCSLib.Online.AdWordsInstance Advertiser Query) → (a : Advertiser) → (a_1 : Query) → self.2 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.AdWordsInstance.budget

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
  {Query : Type u_2} → (self : EconCSLib.Online.AdWordsInstance Advertiser Query) → (a : Advertiser) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.spend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
        (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →
          (a : Advertiser) →
            ∑ q,
              match A q with
              | none => 0
              | some a' => if a' = a then I.bid a q else 0
```

Exact Lean library declaration

```
noncomputable def spend [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (a : Advertiser) : ℝ :=
  ∑ q : Query,
    match A q with
    | none => 0
    | some a' => if a' = a then I.bid a q else 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.CanAssign

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.AdWordsInstance Advertiser Query)
  (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query) (a : Advertiser),
  I.spend A a + I.bid a q ≤ I.budget a
```

Exact Lean library declaration

```
def CanAssign [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (q : Query) (a : Advertiser) : Prop :=
  I.spend A a + I.bid a q ≤ I.budget a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.ChoiceRule

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(Advertiser : Type u_3) →  
(Query : Type u_4) → EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query → Query → Option Advertiser
```

Exact Lean library declaration

```
abbrev ChoiceRule (Advertiser Query : Type*) :=  
  Assignment Advertiser Query → Query → Option Advertiser
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.Feasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.AdWordsInstance Advertiser Query)
  (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) (a : Advertiser), I.spend A a ≤ I.budget a
```

Exact Lean library declaration

```
def Feasible [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) : Prop :=
  ∀ a, I.spend A a ≤ I.budget a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.HistoryState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_3 → Type u_4 → Type (max u_3 u_4)

Exact Lean library declaration

```
structure HistoryState (Advertiser Query : Type*) where
  assignment : Assignment Advertiser Query
  seen : Finset Query
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.HistoryState.assignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} → (self : EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query) → (a : Query) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.HistoryState.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query →  
      Finset Query → EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.HistoryState.seen

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} → (self : EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query) → self.2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.balanceDiscount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(x : \mathbb{R}) \rightarrow 1 - \text{Real.exp } (x - 1)$

Exact Lean library declaration

```
noncomputable def balanceDiscount (x : ℝ) : ℝ :=  
  1 - Real.exp (x - 1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.spentFraction

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
        (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →
          (a : Advertiser) → I.spend A a / I.budget a
```

Exact Lean library declaration

```
noncomputable def spentFraction [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (a : Advertiser) : ℝ :=
  I.spend A a / I.budget a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.balanceScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
        (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →
          (a : Advertiser) →
            (q : Query) → I.bid a q * EconCSLib.Online.AdWordsInstance.balanceDiscount (I.spentFraction A a)
```

Exact Lean library declaration

```
noncomputable def balanceScore [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (a : Advertiser) (q : Query) : ℝ :=
  I.bid a q * balanceDiscount (I.spentFraction A a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.IsBalanceChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [inst : Fintype Query]
  [inst_1 : DecidableEq Advertiser] (I : EconCSLib.Online.AdWordsInstance Advertiser Query)
  (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query) (a : Advertiser),
  I.CanAssign A q a ∧ ∀ (b : Advertiser), I.CanAssign A q b → I.balanceScore A b q ≤ I.balanceScore A a q
```

Exact Lean library declaration

```
def IsBalanceChoice
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (q : Query) (a : Advertiser) : Prop :=
  I.CanAssign A q a ∧
  ∀ b, I.CanAssign A q b → I.balanceScore A b q ≤ I.balanceScore A a q
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.revenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : Fintype Query] →
(I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
(A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →
  ∑ q,
  match A q with
  | none => 0
  | some a => I.bid a q
```

Exact Lean library declaration

```
noncomputable def revenue [Fintype Query]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) : ℝ :=
  ∑ q : Query,
  match A q with
  | none => 0
  | some a => I.bid a q
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.IsOptimalAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.AdWordsInstance Advertiser Query)
  (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query),
  I.Feasible A ∧
  ∀ (B : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query), I.Feasible B → I.revenue B ≤ I.revenue A
```

Exact Lean library declaration

```
def IsOptimalAssignment [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) : Prop :=
  I.Feasible A ∧ ∀ B, I.Feasible B → I.revenue B ≤ I.revenue A
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.NonnegativeBids

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} \{ \text{Query} : \text{Type } u_2 \} (I : \text{EconCSLib.Online.AdWordsInstance Advertiser Query}) (a : \text{Advertiser}) (q : \text{Query}), 0 \leq I.\text{bid } a \ q$

Exact Lean library declaration

`def NonnegativeBids (I : AdWordsInstance Advertiser Query) : Prop :=
 $\forall a \ q, 0 \leq I.\text{bid } a \ q$`

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.NonnegativeBudgets

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} \{ \text{Query} : \text{Type } u_2 \} (I : \text{EconCSLib.Online.AdWordsInstance Advertiser Query}) (a : \text{Advertiser}),$
 $\theta \leq I.\text{budget } a$

Exact Lean library declaration

```
def NonnegativeBudgets (I : AdWordsInstance Advertiser Query) : Prop :=  
   $\forall a, \theta \leq I.\text{budget } a$ 
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.PositiveBudgets

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} \{ \text{Query} : \text{Type } u_2 \} (I : \text{EconCSLib.Online.AdWordsInstance Advertiser Query}) (a : \text{Advertiser}),$
 $\theta < I.\text{budget } a$

Exact Lean library declaration

`def PositiveBudgets (I : AdWordsInstance Advertiser Query) : Prop :=
 $\forall a, \theta < I.\text{budget } a$`

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.assignQuery

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : DecidableEq Query] →  
    (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →  
      (q : Query) → (a : Advertiser) → (a_1 : Query) → if a_1 = q then some a else A a_1
```

Exact Lean library declaration

```
def assignQuery [DecidableEq Query]  
  (A : Assignment Advertiser Query) (q : Query) (a : Advertiser) :  
    Assignment Advertiser Query :=  
  fun q' => if q' = q then some a else A q'
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.exists_balanceChoice_of_exists_canAssign

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Advertiser] [inst_1 : Fintype Query]
  [inst_2 : DecidableEq Advertiser] (I : EconCSLib.Online.AdWordsInstance Advertiser Query)
  (A : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) (q : Query),
  (∃ a, I.CanAssign A q a) → ∃ a, I.IsBalanceChoice A q a
```

Exact Lean library declaration

```
theorem exists_balanceChoice_of_exists_canAssign
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (A : Assignment Advertiser Query) (q : Query)
  (h : ∃ a, I.CanAssign A q a) :
  ∃ a, I.IsBalanceChoice A q a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.balanceChoiceRule

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Advertiser] →
    [inst_1 : Fintype Query] →
      [inst_2 : DecidableEq Advertiser] →
        (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
          (a : EconCSLib.Online.AdWordsInstance.Assignment Advertiser Query) →
            (a_1 : Query) → if h : ∃ a_2, I.CanAssign a a_1 a_2 then some (Classical.choose a_2) else none
```

Exact Lean library declaration

```
noncomputable def balanceChoiceRule
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query) :
  ChoiceRule Advertiser Query := by
  classical
  exact fun A q =>
    if h : ∃ a, I.CanAssign A q a then
      some (Classical.choose (exists_balanceChoice_of_exists_canAssign I A q h))
    else
      none
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.emptyAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} → {Query : Type u_2} → Query → none
```

Exact Lean library declaration

```
def emptyAssignment : Assignment Advertiser Query :=  
  fun _ => none
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.exists_optimalAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [inst : Fintype Query]
  [inst_1 : DecidableEq Advertiser] (I : EconCSLib.Online.AdWordsInstance Advertiser Query),
  I.NonnegativeBudgets → ∃ A, I.IsOptimalAssignment A
```

Exact Lean library declaration

```
theorem exists_optimalAssignment
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]
  (I : AdWordsInstance Advertiser Query)
  (hbudget : I.NonnegativeBudgets) :
  ∃ A : Assignment Advertiser Query, I.IsOptimalAssignment A
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.historyFinset

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Query : Type u_2} →
[inst : DecidableEq Query] →
(a : List Query) →
  List.brecOn a fun x f =>
    (match (motive := (x : List Query) → List.below x → Finset Query) x with
    | [] => fun x => ∅
    | q :: qs => fun x => insert q x.1)
f
```

Exact Lean library declaration

```
def historyFinset [DecidableEq Query] : List Query → Finset Query
| [] => ∅
| q :: qs => insert q (historyFinset qs)

theorem mem_historyFinset [DecidableEq Query]
  {history : List Query} {q : Query} :
  q ∈ historyFinset history ↔ q ∈ history := by
  induction history with
  | nil =>
    simp [historyFinset]
  | cons r rs ih =>
    simp [historyFinset, ih]
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.initialHistoryState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [DecidableEq Query] → { assignment := EconCSLib.Online.AdWordsInstance.emptyAssignment, seen := ∅ }
```

Exact Lean library declaration

```
def initialHistoryState [DecidableEq Query] : HistoryState Advertiser Query where  
  assignment := emptyAssignment  
  seen := ∅
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.maxBidForQuery

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : Fintype Advertiser] →
[inst_1 : Nonempty Advertiser] →
(I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
(q : Query) →
  let bids := Finset.image (fun a => I.bid a q) Finset.univ;
  have hbids := ...;
  max 0 (bids.max' hbids)
```

Exact Lean library declaration

```
noncomputable def maxBidForQuery
  [Fintype Advertiser] [Nonempty Advertiser]
  (I : AdWordsInstance Advertiser Query) (q : Query) : ℝ := by
  classical
  let bids : Finset ℝ :=
    Finset.univ.image fun a : Advertiser => I.bid a q
  have hbids : bids.Nonempty := by
    obtain {a} := (inferInstance : Nonempty Advertiser)
    exact ⟨I.bid a q, by simp [bids]⟩
  exact max 0 (bids.max' hbids)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Advertiser : Type u_1} →  
  {Query : Type u_2} → (Advertiser → ℝ) → (Advertiser → Query → ℝ) → EconCSLib.Online.AdWordsInstance Advertiser Query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.offlineOptimumAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : Fintype Query] →  
      [inst_2 : DecidableEq Advertiser] →  
        (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
          (hbudget : I.NonnegativeBudgets) → (a : Query) → Classical.choose ... a
```

Exact Lean library declaration

```
noncomputable def offlineOptimumAssignment  
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]  
  (I : AdWordsInstance Advertiser Query)  
  (hbudget : I.NonnegativeBudgets) : Assignment Advertiser Query :=  
  Classical.choose (exists_optimalAssignment I hbudget)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.offlineOptimumValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : Fintype Query] →  
      [inst_2 : DecidableEq Advertiser] →  
        (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
          (hbudget : I.NonnegativeBudgets) → I.revenue (I.offlineOptimumAssignment hbudget)
```

Exact Lean library declaration

```
noncomputable def offlineOptimumValue  
  [Fintype Advertiser] [Fintype Query] [DecidableEq Advertiser]  
  (I : AdWordsInstance Advertiser Query)  
  (hbudget : I.NonnegativeBudgets) : ℝ :=  
  I.revenue (I.offlineOptimumAssignment hbudget)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.stepHistoryState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[Fintype Query] →
[DecidableEq Advertiser] →
[inst : DecidableEq Query] →
EconCSLib.Online.AdWordsInstance.Advertiser Query →
(rule : EconCSLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) →
(S : EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query) →
(q : Query) →
  if q ∈ S.seen then { assignment := S.assignment, seen := insert q S.seen }
  else
    match rule S.assignment q with
    | none => { assignment := S.assignment, seen := insert q S.seen }
    | some a =>
      { assignment := EconCSLib.Online.AdWordsInstance.assignQuery S.assignment q a,
        seen := insert q S.seen }
```

Exact Lean library declaration

```
def stepHistoryState
  [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
  (I : AdWordsInstance Advertiser Query)
  (rule : ChoiceRule Advertiser Query)
  (S : HistoryState Advertiser Query) (q : Query) :
  HistoryState Advertiser Query :=
  if q ∈ S.seen then
    { S with seen := insert q S.seen }
  else
    match rule S.assignment q with
    | none => { assignment := S.assignment, seen := insert q S.seen }
    | some a =>
      { assignment := assignQuery S.assignment q a
        seen := insert q S.seen }
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.runHistoryStateFrom

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : Fintype Query] →
[inst_1 : DecidableEq Advertiser] →
[inst_2 : DecidableEq Query] →
(I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
(rule : EconCSLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) →
(a : EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query) →
(a_1 : List Query) →
List.brecOn (motive := fun x =>
  EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query →
  EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query)
a_1
(fun x f x_1 =>
  (match (motive :=
    EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query →
    (x : List Query) →
    List.below (motive := fun x =>
      EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query →
      EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query)
    x →
    EconCSLib.Online.AdWordsInstance.HistoryState Advertiser Query)
    x_1, x with
  | S, [] => fun x => S
  | S, q :: qs => fun x => x.1 (I.stepHistoryState rule S q))
  f)
a
```

Exact Lean library declaration

```
def runHistoryStateFrom
  [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
  (I : AdWordsInstance Advertiser Query)
  (rule : ChoiceRule Advertiser Query) :
  HistoryState Advertiser Query → List Query → HistoryState Advertiser Query
| S, [] => S
| S, q :: qs => runHistoryStateFrom I rule (stepHistoryState I rule S q) qs

/-- Run an online choice rule from the initial empty state. -/
def runHistoryState
  [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
  (I : AdWordsInstance Advertiser Query)
  (rule : ChoiceRule Advertiser Query) (history : List Query) :
  HistoryState Advertiser Query :=
  runHistoryStateFrom I rule initialHistoryState history
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.runHistoryState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      [inst_2 : DecidableEq Query] →
        (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
          (rule : EconCSLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) →
            (history : List Query) →
              I.runHistoryStateFrom rule EconCSLib.Online.AdWordsInstance.initialHistoryState history
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.runAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      [inst_2 : DecidableEq Query] →
        (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →
          (rule : EconCSLib.Online.AdWordsInstance.ChoiceRule Advertiser Query) →
            (history : List Query) → (a : Query) → (I.runHistoryState rule history).assignment a
```

Exact Lean library declaration

```
def runAssignment
  [Fintype Query] [DecidableEq Advertiser] [DecidableEq Query]
  (I : AdWordsInstance Advertiser Query)
  (rule : ChoiceRule Advertiser Query) (history : List Query) :
  Assignment Advertiser Query :=
  (runHistoryState I rule history).assignment
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.withEffectiveBids

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
  {Query : Type u_2} →  
    (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
      (effectiveBid : Advertiser → Query → ℝ) → { budget := I.budget, bid := effectiveBid }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.withAdvertiserWeights

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
(I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
(weight : Advertiser → ℝ) → I.withEffectiveBids fun a q => weight a * I.bid a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.withAvailability

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
    (active : Advertiser → Query → Prop) →  
      [inst : (a : Advertiser) → (q : Query) → Decidable (active a q)] →  
        I.withEffectiveBids fun a q => if active a q then I.bid a q else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.AdWordsInstance.withClickThroughRates

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
(I : EconCSLib.Online.AdWordsInstance Advertiser Query) →  
(ctr : Advertiser → Query → ℝ) → I.withEffectiveBids fun a q => ctr a q * I.bid a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.theorem9BidderSpendUpperBound

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (\text{bidder} : \text{Fin } N) \rightarrow \min 1 (\sum \text{round}, \text{if } \uparrow \text{round} \leq \uparrow \text{bidder} \text{ then } 1 / \uparrow (N - \uparrow \text{round}) \text{ else } 0)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.theorem9NormalizedRevenueUpperBound

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (\sum \text{bidder}, \text{EconCSLib.Online.theorem9BidderSpendUpperBound } N \text{ bidder}) / \uparrow N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.theorem9EligibleBidders

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (\text{round} : \text{Fin } N) \rightarrow \{\text{bidder} \mid \uparrow \text{round} \leq \uparrow \text{bidder}\}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.theorem9ActualEligibleBidders

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (permutation : Equiv.Perm (Fin N)) →  
    (round : Fin N) → Finset.image (↑permutation) (EconCSLib.Online.theorem9EligibleBidders N round)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.theorem9ObservedPrefix

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (permutation : Equiv.Perm (Fin N)) →  
    (currentRound query : Fin N) →  
      if ↑query ≤ ↑currentRound then EconCSLib.Online.theorem9ActualEligibleBidders N permutation query else ∅
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(prefixAllocation : Algorithm → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  normalizedRevenue algorithm permutation ≤
    (∑ bidder,
      min 1
        (∑ round,
          prefixAllocation algorithm (EconCSLib.Online.theorem90observedPrefix N permutation round) round
            (permutation bidder))) /
    ↑N) →
(∀ (algorithm : Algorithm) (obs : Fin N → Finset (Fin N)) (round bidder : Fin N),
  bidder ∉ obs round → prefixAllocation algorithm obs round bidder = 0) →
(∀ (algorithm : Algorithm) (obs : Fin N → Finset (Fin N)) (round : Fin N),
  ∑ bidder ∈ obs round, prefixAllocation algorithm obs round bidder ≤ 1) →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.prefixAllocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
  (self : EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →
    (a : Algorithm) → (a_1 : Fin N → Finset (Fin N)) → (a_2 a_3 : Fin N) → self.2 a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.normalizedRevenue_le_cappedPrefixAllocationSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
    (∑ bidder,
      min 1
        (∑ round,
          self.prefixAllocation algorithm (EconCSLib.Online.theorem90bservedPrefix N permutation round) round
            (permutation bidder))) /
    ↑N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate.revenueBound_le_ratio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio),
  EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(allocation : Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ) →
(prefixAllocation : Algorithm → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  allocation algorithm permutation round bidder =
  prefixAllocation algorithm (EconCSLib.Online.theorem9ObservedPrefix N permutation round) round
  (permutation bidder)) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  normalizedRevenue algorithm permutation ≤
  (∑ bidder, min 1 (∑ round, allocation algorithm permutation round bidder)) / ↑N) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  ¬↑round ≤ ↑bidder → allocation algorithm permutation round bidder = 0) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
  allocation algorithm permutation round bidder ≤
  1) →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate. toObservedPrefixAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(C : EconCSLib.Online.BMatchingFeasibleObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →
{ normalizedRevenue := C.normalizedRevenue,
  allocation := fun algorithm permutation round bidder =>
    C.prefixAllocation algorithm (EconCSLib.Online.theorem90observedPrefix N permutation round) round
    (permutation bidder),
  prefixAllocation := C.prefixAllocation, allocation_eq_prefix := "",
  normalizedRevenue_le_cappedAllocationSpend := "", allocation_zero_of_ineligible := "",
  round_allocation_sum_le_one := "", revenueBound_le_ratio := "" }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (\text{Fin } N \rightarrow \text{Finset } (\text{Fin } N)) \rightarrow \text{Fin } N \rightarrow \text{Option } (\text{Fin } N)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixChoice.Feasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{N : \mathbb{N}\} \text{ (choice : EconCSLib.Online.BMatchingIntegralPrefixChoice } N) \text{ (obs : Fin } N \rightarrow \text{Finset (Fin } N))$
 $\text{(round bidder : Fin } N), \text{ choice obs round = some bidder} \rightarrow \text{bidder} \in \text{obs round}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \{ \text{choice} \ // \ \text{choice.Feasible} \}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm N) →  
  (obs : Fin N → Finset (Fin N)) → (round bidder : Fin N) → if !algorithm obs round = some bidder then 1 else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_sum_le_one

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\forall \{N : \mathbb{N}\} \text{ (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm } N) \text{ (obs : Fin } N \rightarrow \text{Finset (Fin } N))$
 $\text{ (round : Fin } N), \sum \text{ bidder } \in \text{ obs round, algorithm.prefixAllocation obs round bidder } \leq 1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation_zero_of_not_visible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\forall \{N : \mathbb{N}\} \text{ (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm } N) \text{ (obs : Fin } N \rightarrow \text{Finset (Fin } N))$
 $\text{ (round bidder : Fin } N), \text{ bidder} \notin \text{obs round} \rightarrow \text{algorithm.prefixAllocation obs round bidder} = 0$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.allocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → (a_2 a_3 : Fin N) → self.2 a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.allocation_zero_of_ineligible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  → round ≤ bidder → self.allocation algorithm permutation round bidder = 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
    (∑ bidder, min 1 (∑ round, self.allocation algorithm permutation round bidder)) / ↑N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.revenueBound_le_ratio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio),
  EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.round_allocation_sum_le_one

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round, self.allocation algorithm permutation round bidder ≤ 1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(allocation : Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  normalizedRevenue algorithm permutation ≤
    (∑ bidder, min 1 (∑ round, allocation algorithm permutation round bidder)) / ↑N) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  ¬↑round ≤ ↑bidder → allocation algorithm permutation round bidder = 0) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
    allocation algorithm permutation round bidder ≤ 1) →
(relabelInput : Fin N → Fin N → Fin N → Equiv.Perm (Fin N) = Equiv.Perm (Fin N)) →
(∀ (algorithm : Algorithm) (round bidder bidder' : Fin N),
  ↑round ≤ ↑bidder →
  ↑round ≤ ↑bidder' →
  ∀ (permutation : Equiv.Perm (Fin N)),
    allocation algorithm ((relabelInput round bidder bidder') permutation) round bidder =
      allocation algorithm permutation round bidder') →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

**EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate.
toRelabelSymmetricPointwiseAllocationRevenueCertificate**

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(C : EconCSLib.Online.BMatchingObservedPrefixAllocationRevenueCertificate N Algorithm ratio) →
{ normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,
  normalizedRevenue_le_cappedAllocationSpend := "", allocation_zero_of_ineligible := "",
  round_allocation_sum_le_one := "",
  relabelInput := fun x bidder bidder' => Equiv.mulRight (Equiv.swap bidder bidder'),
  allocation_eq_of_relabel_eligible := "", revenueBound_le_ratio := "" }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.allocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
{Algorithm : Type u_1} →  
{ratio : ℝ} →  
  (self : EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
    (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → (a_2 a_3 : Fin N) → self.2 a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.uniformPMF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\alpha : \text{Type } u_1) \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{inst}_1 : \text{Nonempty } \alpha] \rightarrow \text{PMF.ofFintype } (\text{fun } x \Rightarrow (\uparrow(\text{Fintype.card } \alpha))^{-1}) \dots$

Exact Lean library declaration

```
noncomputable def uniformPMF (α : Type*) [Fintype α] [Nonempty α] : PMF α :=
  PMF.ofFintype (fun _ : α => ((Fintype.card α : ENNReal)⁻¹)) (by
    rw [Finset.sum_const, Finset.card_univ]
    have hcard : (Fintype.card α : ENNReal) ≠ 0 := by
      exact_mod_cast (Fintype.card_pos_iff.mpr ⟨Nonempty α⟩.ne')
    simp [nsmul_eq_mul] using
      (ENNReal.mul_inv_cancel hcard (ENNReal.natCast_ne_top (Fintype.card α))))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.uniformPermutationDistribution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.uniformPMF } (\text{Equiv.Perm } (\text{Fin } N))$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{DecidableEq } \alpha] \rightarrow (\mu : \text{PMF } \alpha) \rightarrow (f : \alpha \rightarrow \mathbb{R}) \rightarrow \sum a, (\mu a).toReal * f a$

Exact Lean library declaration

```
noncomputable def pmfExp {α : Type*} [Fintype α] [DecidableEq α]
  (μ : PMF α) (f : α → ℝ) : ℝ :=
  ∑ a, (μ a).toReal * f a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.expectedAllocation_le

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio) (algorithm : Algorithm)
  (round bidder : Fin N),
  (EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
    self.allocation algorithm permutation round bidder) ≤
    if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(allocation : Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  normalizedRevenue algorithm permutation ≤
  (∑ bidder, min 1 (∑ round, allocation algorithm permutation round bidder)) / ↑N) →
(∀ (algorithm : Algorithm) (round bidder : Fin N),
  (EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
    allocation algorithm permutation round bidder) ≤
    if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0) →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.revenueBound_le_ratio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio),
  EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(expectedRoundBidderAllocation : Algorithm → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm),
  (EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
    normalizedRevenue algorithm permutation) ≤
    (∑ bidder, min 1 (∑ round, expectedRoundBidderAllocation algorithm round bidder)) / ↑N) →
(∀ (algorithm : Algorithm) (round bidder : Fin N),
  expectedRoundBidderAllocation algorithm round bidder ≤
    if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0) →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(C : EconCSLib.Online.BMatchingPointwiseAllocationRevenueCertificate N Algorithm ratio) →
{ normalizedRevenue := C.normalizedRevenue,
  expectedRoundBidderAllocation := fun algorithm round bidder =>
    EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
      C.allocation algorithm permutation round bidder,
  deterministicAverage_le_cappedExpectedSpend := ∞, expectedRoundBidderAllocation_le := ∞,
  revenueBound_le_ratio := ∞ }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.allocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
  (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →
    (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → (a_2 a_3 : Fin N) → self.2 a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.allocation_zero_of_ineligible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  ¬round ≤ round bidder → self.allocation algorithm permutation round bidder = 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
    (∑ bidder, min 1 (∑ round, self.allocation algorithm permutation round bidder)) / ↑N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.revenueBound_le_ratio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio),
  EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate.round_allocation_sum_le_one

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round, self.allocation algorithm permutation round bidder ≤ 1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \text{Type } u_1 \rightarrow \mathbb{R} \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(normalizedRevenue : Algorithm → Equiv.Perm (Fin N) → ℝ) →
(allocation : Algorithm → Equiv.Perm (Fin N) → Fin N → Fin N → ℝ) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  normalizedRevenue algorithm permutation ≤
  (∑ bidder, min 1 (∑ round, allocation algorithm permutation round bidder)) / ↑N) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round bidder : Fin N),
  ¬↑round ≤ ↑bidder → allocation algorithm permutation round bidder = 0) →
(∀ (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
  allocation algorithm permutation round bidder ≤
  1) →
(∀ (algorithm : Algorithm) (round bidder bidder' : Fin N),
  ↑round ≤ ↑bidder →
  ↑round ≤ ↑bidder' →
  (EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
    allocation algorithm permutation round bidder) =
    EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
    allocation algorithm permutation round bidder') →
EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio →
EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate. toSymmetricPointwiseAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →
{Algorithm : Type u_1} →
{ratio : ℝ} →
(C : EconCSLib.Online.BMatchingRelabelSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →
{ normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,
  normalizedRevenue_le_cappedAllocationSpend := ..., allocation_zero_of_ineligible := ...,
  round_allocation_sum_le_one := ..., expectedAllocation_eq_of_eligible := ..., revenueBound_le_ratio := ... }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingRoundAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.allocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → (a_2 a_3 : Fin N) → self.2 a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (self : EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
        (a : Algorithm) → (a_1 : Equiv.Perm (Fin N)) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.normalizedRevenue_le_cappedAllocationSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio)
  (algorithm : Algorithm) (permutation : Equiv.Perm (Fin N)),
  self.normalizedRevenue algorithm permutation ≤
    (∑ bidder, min 1 (∑ round, self.allocation algorithm permutation round bidder)) / ↑N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate.revenueBound_le_ratio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {N : ℕ} {Algorithm : Type u_1} {ratio : ℝ}
  (self : EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio),
  EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ≤ ratio
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate. toPointwiseAllocationRevenueCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{N : ℕ} →  
  {Algorithm : Type u_1} →  
    {ratio : ℝ} →  
      (C : EconCSLib.Online.BMatchingSymmetricPointwiseAllocationRevenueCertificate N Algorithm ratio) →  
        { normalizedRevenue := C.normalizedRevenue, allocation := C.allocation,  
          normalizedRevenue_le_cappedAllocationSpend := ..., expectedAllocation_le := ..., revenueBound_le_ratio := ... }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$(\text{Algorithm} : \mathbb{N} \rightarrow \text{Type } u_1) \rightarrow [(\mathbb{N} : \mathbb{N}) \rightarrow \text{Fintype } (\text{Algorithm } N)] \rightarrow [(\mathbb{N} : \mathbb{N}) \rightarrow \text{DecidableEq } (\text{Algorithm } N)] \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Algorithm : ℕ → Type u_1} →
[inst : (N : ℕ) → Fintype (Algorithm N)] →
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →
(prefixAllocation : (N : ℕ) → Algorithm N → (Fin N → Finset (Fin N)) → Fin N → Fin N → ℝ) →
(∀ (N : ℕ) (algorithm : Algorithm N) (obs : Fin N → Finset (Fin N)) (round bidder : Fin N),
bidder ∉ obs round → prefixAllocation N algorithm obs round bidder = 0) →
(∀ (N : ℕ) (algorithm : Algorithm N) (obs : Fin N → Finset (Fin N)) (round : Fin N),
∑ bidder ∈ obs round, prefixAllocation N algorithm obs round bidder ≤ 1) →
EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily Algorithm
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.prefixAllocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Algorithm : ℕ → Type u_1} →  
[inst : (N : ℕ) → Fintype (Algorithm N)] →  
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →  
(self : EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily Algorithm) →  
(N : ℕ) → (a : Algorithm N) → (a_1 : Fin N → Finset (Fin N)) → (a_2 a_3 : Fin N) → self.1 N a a_1 a_2 a_3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Algorithm : ℕ → Type u_1} →
[inst : (N : ℕ) → Fintype (Algorithm N)] →
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →
(C : EconCSLib.Online.BMatchingTheorem9FeasiblePrefixRuleFamily Algorithm) →
(N : ℕ) →
  (algorithm : Algorithm N) →
  (permutation : Equiv.Perm (Fin N)) →
  (∑ bidder,
    min 1
      (∑ round,
        C.prefixAllocation N algorithm (EconCSLib.Online.theorem90bservedPrefix N permutation round) round
        (permutation bidder))) /
  ↑N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$(\text{Algorithm} : \mathbb{N} \rightarrow \text{Type } u_1) \rightarrow [(\mathbb{N} : \mathbb{N}) \rightarrow \text{Fintype } (\text{Algorithm } N)] \rightarrow [(\mathbb{N} : \mathbb{N}) \rightarrow \text{DecidableEq } (\text{Algorithm } N)] \rightarrow \text{Type } u_1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.choice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Algorithm : ℕ → Type u_1} →  
[inst : (N : ℕ) → Fintype (Algorithm N)] →  
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →  
(self : EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily Algorithm) →  
(N : ℕ) → (a : Algorithm N) → (a_1 : Fin N → Finset (Fin N)) → (a_2 : Fin N) → self.1 N a a_1 a_2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Algorithm : ℕ → Type u_1} →  
  [inst : (N : ℕ) → Fintype (Algorithm N)] →  
    [inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →  
      (choice : (N : ℕ) → Algorithm N → EconCSLib.Online.BMatchingIntegralPrefixChoice N) →  
        (∀ (N : ℕ) (algorithm : Algorithm N), (choice N algorithm).Feasible) →  
          EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily Algorithm
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.toFeasiblePrefixRuleFamily

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Algorithm : ℕ → Type u_1} →  
[inst : (N : ℕ) → Fintype (Algorithm N)] →  
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →  
(C : EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily Algorithm) →  
{  
  prefixAllocation := fun N algorithm obs round bidder =>  
    if C.choice N algorithm obs round = some bidder then 1 else 0,  
  prefixAllocation_zero_of_not_visible := ..., prefixAllocation_sum_le_one := ... }  
}
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily.normalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Algorithm : ℕ → Type u_1} →  
[inst : (N : ℕ) → Fintype (Algorithm N)] →  
[inst_1 : (N : ℕ) → DecidableEq (Algorithm N)] →  
(C : EconCSLib.Online.BMatchingTheorem9IntegralPrefixChoiceFamily Algorithm) →  
(N : ℕ) →  
  (algorithm : Algorithm N) →  
    (permutation : Equiv.Perm (Fin N)) → C.toFeasiblePrefixRuleFamily.normalizedRevenue N algorithm permutation
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedIndividualBudget

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\mathbb{R} \rightarrow 1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (spend : ℝ → ℝ),
  IntervalIntegrable spend MeasureTheory.volume 0 1 ∧
    ∀ bidder ∈ Set.Icc 0 1,
      0 ≤ spend bidder ∧ spend bidder ≤ EconCSLib.Online.MSVV07Appendix.printedIndividualBudget bidder
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaU

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \kappa^{-1}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaY

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07Appendix.derivedKappaU } \kappa ^ 2$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaZ

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \text{Real.exp } (1 - \text{EconCSLib.Online.MSVV07Appendix.derivedKappaU } \kappa)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07Appendix.derivedKappaZ } \kappa + \text{EconCSLib.Online.MSVV07Appendix.derivedKappaY } \kappa$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaB

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow$
(EconCSLib.Online.MSVV07Appendix.derivedKappaZ κ - 1) / EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator κ

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaT

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow (\text{EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator } \kappa)^{-1}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaX

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07Appendix.derivedKappaY } \kappa / \text{EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator } \kappa$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(κ : ℝ) →
EconCSLib.Online.MSVV07Appendix.derivedKappaB κ + EconCSLib.Online.MSVV07Appendix.derivedKappaX κ +
EconCSLib.Online.MSVV07Appendix.derivedKappaT κ

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\kappa : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07Appendix.derivedKappaB } \kappa + \kappa * \text{EconCSLib.Online.MSVV07Appendix.derivedKappaX } \kappa$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.derivedKappaRatio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(κ : ℝ) →  
(EconCSLib.Online.MSVV07Appendix.derivedKappaZ κ - 1 + EconCSLib.Online.MSVV07Appendix.derivedKappaU κ) /  
EconCSLib.Online.MSVV07Appendix.derivedKappaDenominator κ
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.instDecidable_mSVV07AdWords

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(p : \text{Prop}) \rightarrow \text{Classical.propDecidable } p$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\int (_bidder : \mathbb{R}) \text{ in } 0..1, 1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

3 / 5 - Real.log (9 / 5)

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseThreeServedMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

1 / 2 * EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07Appendix.printedContinuousOnlineValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$2 / 5 + 2 * (1 / 10) + \text{EconCSLib.Online.MSVV07Appendix.printedPhaseThreeServedMass}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(r : \mathbb{R}) \mapsto (9 / 10 - r)^{-1}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseOneEligible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (r \text{ bidder} : \mathbb{R}), 1 / 10 + r \leq \text{bidder} \wedge \text{bidder} \leq 1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseOneRawBid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(r bidder : $\mathbb{R}$) → if EconCSLib.Online.MSVV07Appendix.printedPhaseOneEligible r bidder then 1 else 0

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(r bidder : ℝ) →  
  ∫ (s : ℝ) in 0..r,  
    if EconCSLib.Online.MSVV07Appendix.printedPhaseOneEligible s bidder then  
      EconCSLib.Online.MSVV07Appendix.printedPhaseOneAllocationRate s  
    else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(progress owner bidder : ℝ) →  
  if 0 ≤ progress ∧ progress ≤ EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail ∧ bidder = owner then 1  
  else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseThreeRawBid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(owner bidder : $\mathbb{R}$) → if bidder = owner then 1 else 0

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\text{bidder} : \mathbb{R}) \rightarrow \text{if } 1 / 2 \leq \text{bidder} \wedge \text{bidder} \leq 1 \text{ then } 2 \text{ else } 0$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseTwoRawBid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(offset bidder : $\mathbb{R}$) → if $1 / 2 \leq \text{bidder} \wedge \text{bidder} \leq 1$ then 2 else if bidder = offset then 1 else 0

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(offset bidder : ℝ) →  
  EconCSLib.Online.MSVV07Appendix.printedPhaseOneSpentBefore (2 / 5) bidder +  
  ⌈ (s : ℝ) in 0..offset,  
    EconCSLib.Online.MSVV07Appendix.printedPhaseTwoRawBid s bidder *  
    EconCSLib.Online.MSVV07Appendix.printedPhaseTwoAllocationDensity bidder
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(progress owner : ℝ) →  
  EconCSLib.Online.MSVV07Appendix.printedPhaseTwoSpentBefore (1 / 10) owner +  
  ⌈ (s : ℝ) in 0..progress,  
    EconCSLib.Online.MSVV07Appendix.printedPhaseThreeAllocationRate s owner owner *  
    EconCSLib.Online.MSVV07Appendix.printedPhaseThreeRawBid owner owner
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_1) → (Query : Type u_2) → Query → Option Advertiser

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.PaperAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_1) → (Query : Type u_2) → EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_1) → (Query : Type u_2) → EconCSLib.Online.AdWordsInstance Advertiser Query

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.PaperInstance

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_1) → (Query : Type u_2) → EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.instDecidable_mSVV07AdWords

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(p : \text{Prop}) \rightarrow \text{Classical.propDecidable } p$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(k : \mathbb{N}) \rightarrow (i : \text{Fin } k) \rightarrow 1 - \text{Real.exp } (-(1 - \uparrow(i + 1) / \uparrow k))$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

`(k : ℕ) → (i : Fin k) → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff k i`

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(c \text{ epsilon} : \mathbb{R}) \rightarrow (c + \text{epsilon}) / (2 * c + \text{epsilon})$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$$\forall \{k : \mathbb{N}\} (\text{alphaTotal} : \mathbb{R}) (\text{alpha beta} : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$$
$$(\text{alphaTotal} - \sum_{j \in \text{Finset.Iic } i, \text{ alpha } j}) / \uparrow k \leq \text{beta } i$$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{k : \mathbb{N}\} (\text{psi alpha beta} : \text{Fin } k \rightarrow \mathbb{R}), \sum i, \text{psi } i * \text{alpha } i \leq \sum i, \text{psi } i * \text{beta } i$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) →
        (A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) →
          (a : Advertiser) →
            ∑ q,
              match A q with
              | none => 0
              | some a' => if a' = a then I.bid a q else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperTradeoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(s : \mathbb{R}) \rightarrow 1 - \text{Real.exp } (s - 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperBalanceScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Query] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) →
        (A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) →
          (a : Advertiser) →
            (q : Query) →
              I.bid a q *
                EconCSLib.Online.MSVV07PaperFacing.Proof.paperTradeoff
                  (EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend I A a / I.budget a)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperCanAssign

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) (q : Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend I A a + I.bid a q ≤ I.budget a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend I A a ≤ I.budget a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperIsBalanceChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [Fintype Advertiser] [inst : Fintype Query]
  [inst_1 : DecidableEq Advertiser] (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) (q : Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperCanAssign I A q a ∧
    ∀ (b : Advertiser),
      EconCSLib.Online.MSVV07PaperFacing.Proof.paperCanAssign I A q b →
        EconCSLib.Online.MSVV07PaperFacing.Proof.paperBalanceScore I A b q ≤
          EconCSLib.Online.MSVV07PaperFacing.Proof.paperBalanceScore I A a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

1 - 1 / Real.exp 1

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.Proof.paperRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : Fintype Query] →
(I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) →
(A : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperAssignment Advertiser Query) →
  ∑ q,
  match A q with
  | none => 0
  | some a => I.bid a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.paperSmallBids

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2}
  (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) (epsilon : ℝ) (a : Advertiser)
  (q : Query), I.bid a q ≤ epsilon * I.budget a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : Fintype Advertiser] →
[inst_1 : DecidableEq Advertiser] →
(I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) →
(alive : Advertiser → Query → Prop) →
[inst_2 : (a : Advertiser) → (q : Query) → Decidable (alive a q)] →
(a : Advertiser) →
(q : Query) →
  have others := {b ∈ Finset.univ.erase a | alive b q};
  if h : others.Nonempty then max 0 (others.sup' h fun b => I.bid b q) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.Proof.PaperInstance Advertiser Query) →  
        (a : Advertiser) →  
          (q : Query) →  
            have others := Finset.univ.erase a;  
            if h : others.Nonempty then max 0 (others.sup' h fun b => I.bid b q) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9IntegralPrefixAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.Online.BMatchingIntegralPrefixAlgorithm } N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.instDecidableEqBMatchingIntegralPrefixChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (a\ b : \text{EconCSLib.Online.BMatchingIntegralPrefixChoice } N) \rightarrow \text{inferInstance } a\ b$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.instDecidableEqBMatchingIntegralPrefixAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (a\ b : \text{EconCSLib.Online.BMatchingIntegralPrefixAlgorithm } N) \rightarrow \text{inferInstance } a\ b$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.instDecidablePredBMatchingIntegralPrefixChoiceFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (a : EconCSLib.Online.BMatchingIntegralPrefixChoice N) →  
    Classical.decPred EconCSLib.Online.BMatchingIntegralPrefixChoice.Feasible a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening **Verdict:** not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.instFintypeBMatchingIntegralPrefixChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{inferInstance}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.instFintypeBMatchingIntegralPrefixAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{inferInstance}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.paper_adwords_theorem9_integral_prefix_algorithm_family

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ choice := fun _N algorithm => ralgorithm,  
  choice_feasible := EconCSLib.Online.paper_adwords_theorem9_integral_prefix_algorithm_family._proof_1 }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9CappedNormalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (algorithm : EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9IntegralPrefixAlgorithm N) →  
  (permutation : Equiv.Perm (Fin N)) →  
    EconCSLib.Online.paper_adwords_theorem9_integral_prefix_algorithm_family.normalizedRevenue N algorithm permutation
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.Online.uniformPermutationDistribution } N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9RandomizedOnlineAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{PMF } (\text{EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9IntegralPrefixAlgorithm } N)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Advertiser} : \text{Type } u_3 \} \{ \text{Query} : \text{Type } u_4 \} (I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query})$
 $(a : \text{Advertiser}), I.\text{Budget } a = 1$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\mathbb{N} \rightarrow \mathbb{R} \rightarrow \text{Type}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.activeMassBeforeSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
{N : ℝ} →
  (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N) → (a : ℕ) → self.2 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization. allocationPerActiveBidder

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
{N : ℝ} →  
  (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N) → (a : ℕ) → self.4 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.fullMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow \{N : \mathbb{R}\} \rightarrow (\text{self} : \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization } m \ N) \rightarrow \text{self}.6$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow (i : \text{Fin } m) \rightarrow \uparrow(i + 1) / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteCandidateValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{R}) \rightarrow (k : \mathbb{N}) \rightarrow N * (1 - 1 / \uparrow k) ^ k$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow (\mathbb{N} : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07SourceLemmas.paperRouteCandidateValue } \mathbb{N} \ (m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteCoeff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i\ j : \text{Fin } m) \rightarrow 1 + (\uparrow i - \uparrow j) / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i : \text{Fin } m) \rightarrow \{j \mid \uparrow j \leq \uparrow i\}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteLProW

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
(x : Fin m → ℝ) →  
(i : Fin m) →  
  ∑ j ∈ EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix i,  
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteCoeff i j * x j
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (N : \mathbb{R}) \rightarrow (i : \text{Fin } m) \rightarrow N / \uparrow(m + 1) * (1 - 1 / \uparrow(m + 1)) ^ \uparrow i$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (N : \mathbb{R}) \rightarrow (i : \text{Fin } m) \rightarrow \uparrow(i + 1) / \uparrow(m + 1) * N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{m : ℕ} →
{N : ℝ} →
(typeMass : Fin m → ℝ) →
(activeMassBeforeSlab spentBeforeSlab allocationPerActiveBidder queryMassAtSlab : ℕ → ℝ) →
(fullMass revenue : ℝ) →
(∀ (i : Fin m), typeMass i = EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate N i) →
(∀ (i : Fin m), activeMassBeforeSlab i - activeMassBeforeSlab (i + 1) = typeMass i) →
(∀ stage ≤ m, spentBeforeSlab stage = i stage / i(m + 1)) →
(∀ stage ≤ m, allocationPerActiveBidder stage = 1 / i(m + 1)) →
(∀ stage ≤ m, queryMassAtSlab stage = activeMassBeforeSlab stage / i(m + 1)) →
(∀ stage < m,
  spentBeforeSlab (stage + 1) = spentBeforeSlab stage + allocationPerActiveBidder stage) →
(∀ stage ≤ m, spentBeforeSlab stage + allocationPerActiveBidder stage ≤ 1) →
(∀ stage ≤ m,
  activeMassBeforeSlab stage * allocationPerActiveBidder stage = queryMassAtSlab stage) →
  ∑ i, typeMass i + fullMass = N →
  (∀ (i : Fin m),
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow typeMass i =
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i) →
    revenue =
      ∑ i,
        typeMass i *
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction
          m i +
        fullMass →
    revenue =
      ∑ stage ∈ Finset.range (m + 1),
        activeMassBeforeSlab stage * allocationPerActiveBidder stage →
    revenue = N - EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue m N →
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.queryMassAtSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
{N : ℝ} →
  (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N) → (a : ℕ) → self.5 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.revenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow \{N : \mathbb{R}\} \rightarrow (\text{self} : \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization } m \ N) \rightarrow \text{self}.7$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.spentBeforeSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
{N : ℝ} →
  (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N) → (a : ℕ) → self.3 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization.typeMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
{N : ℝ} →  
  (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FactorLPTightFluidRealization m N) → (a : Fin m) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_3 → Type u_3

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.candidateTests

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
(self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser) → self.2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Advertiser : Type u_3} →  
  Option Advertiser → ℕ → ℕ → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.scoreComparisons

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
(self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser) → self.3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult.winner

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
(self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser) → self.1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

N → Type

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution.finalSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{N : \mathbb{N}\} \rightarrow (\text{self} : \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution } N) \rightarrow (a : \text{Fin } N) \rightarrow \text{self}.3 \ a$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{N : ℕ} →
  (allocation spentBefore : Fin N → Fin N → ℝ) →
  (finalSpend : Fin N → ℝ) →
  (∀ (round bidder : Fin N), ¬↑round ≤ ↑bidder → allocation round bidder = 0) →
  (∀ (round bidder bidder' : Fin N),
    ↑round ≤ ↑bidder → ↑round ≤ ↑bidder' → spentBefore round bidder = spentBefore round bidder') →
  (∀ (round bidder : Fin N), 0 ≤ allocation round bidder) →
  (∀ (round bidder : Fin N), spentBefore round bidder + allocation round bidder ≤ 1) →
  (∀ (round bidder : Fin N),
    spentBefore round bidder = ∑ previous, if ↑previous < ↑round then allocation previous bidder else 0) →
  (∀ (round next bidder : Fin N),
    ↑next = ↑round + 1 → spentBefore next bidder = spentBefore round bidder + allocation round bidder) →
  (∀ (round : Fin N),
    ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round, allocation round bidder ≤ 1) →
  (∀ (round : Fin N),
    ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round, allocation round bidder = 1 ∧
    ∀ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
      spentBefore round bidder + allocation round bidder = 1) →
  (∀ (bidder : Fin N), finalSpend bidder = ∑ round, allocation round bidder) →
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FluidBalanceExecution N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsAlphaCompetitive

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Instance} : \text{Type } u_3 \} \text{ (onlineRevenue offlineRevenue : Instance} \rightarrow \mathbb{R}) \text{ (}\alpha : \mathbb{R}) \text{ (I : Instance),}$
 $\alpha \leq \text{onlineRevenue } \bar{I} / \text{offlineRevenue } I$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$$\forall (k : \mathbb{N}) (\text{spentFraction} : \mathbb{R}) (i : \text{Fin } k),$$
$$\text{spentFraction} = 0 \wedge \dagger i = 0 \vee \dagger i / \dagger k < \text{spentFraction} \wedge \text{spentFraction} \leq \dagger(\dagger i + 1) / \dagger k$$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (weight spent fairShare nextWeight : ℝ),
  spent < fairShare ∧ weight < nextWeight →
    fairShare < spent ∧ nextWeight < weight → spent = fairShare ∧ nextWeight = weight
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (a : ℝ) (a_1 : List (ℝ × ℝ)) (a_2 : ℝ),
  List.brecOn (motive := fun x => ℝ → ℝ → Prop) a_1
    (fun x f x_1 x_2 =>
      (match (motive := ℝ → (x : List (ℝ × ℝ)) → ℝ → List.below (motive := fun x => ℝ → ℝ → Prop) x → Prop) x_1, x,
        x_2 with
      | weight, [], finalWeight => fun x => finalWeight = weight
      | weight, (spent, fairShare) :: windows, finalWeight => fun x =>
        ∃ nextWeight,
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment weight spent fairShare nextWeight ∧
            x.1 nextWeight finalWeight)
    f)
  a a_2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment._f

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (x : List (ℝ × ℝ)) (f : List.below (motive := fun x => ℝ → ℝ → Prop) x) (x_1 x_2 : ℝ),
  (match (motive := ℝ → (x : List (ℝ × ℝ)) → ℝ → List.below (motive := fun x => ℝ → ℝ → Prop) x → Prop) x_1, x, x_2 with
  | weight, [], finalWeight => fun x => finalWeight = weight
  | weight, (spent, fairShare) :: windows, finalWeight => fun x =>
    ∃ nextWeight,
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment weight spent fairShare nextWeight ∧
        x.1 nextWeight finalWeight)
  f
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type $u_3 \rightarrow \text{Type } u_4 \rightarrow \text{Type } (\max u_3 u_4)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    Query →  
      Option Advertiser →  
        (Advertiser → ℝ) → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.query

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) → self.1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.spentBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) →  
      (a : Advertiser) → self.3 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision.winner

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) → self.2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_3 → Type u_4 → Type (max u_3 u_4)

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.candidateTests

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) → self.4
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.decisionsRev

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) → self.3
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (Advertiser → ℝ) →  
      ℝ →  
        List (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) →  
          ℕ → ℕ → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.revenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) → self.2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.scoreComparisons

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) → self.5
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState.spent

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →  
      (a : Advertiser) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(Advertiser : Type u_3) → (m : ℕ) → Advertiser × Fin m

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_3 → Type u_4 → Type u_5 → Type (max (max u_3 u_4) u_5)

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Time : Type u_3} →  
  {Regime : Type u_4} →  
    {Query : Type u_5} →  
      (Time → Regime) →  
        (Regime → PMF Query) →  
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistribution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Time : Type u_3} →  
  {Regime : Type u_4} →  
    {Query : Type u_5} →  
      (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query) →  
        (a : Regime) → self.2 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.regimeAt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Time : Type u_3} →  
  {Regime : Type u_4} →  
    {Query : Type u_5} →  
      (self : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query) →  
        (a : Time) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution.queryDistributionAt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Time : Type u_3} →  
  {Regime : Type u_4} →  
    {Query : Type u_5} →  
      (M : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query) →  
        (time : Time) → M.queryDistribution (M.regimeAt time)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(k : \mathbb{N}) \rightarrow (hk : 0 < k) \rightarrow (\text{spentFraction} : \mathbb{R}) \rightarrow (\min [\max 0 (\uparrow k * \text{spentFraction})] + (k - 1), \neg)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(k : \mathbb{N}) \rightarrow$
 $(hk : 0 < k) \rightarrow (\text{spent budget} : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab } k \text{ } hk \text{ } (\text{spent} / \text{budget})$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : DecidableEq Advertiser] →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(q : Query) →
(scan : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser) →
have charge := fun a => I.bid a q;
{
  spent := fun a =>
    match scan.winner with
    | none => S.spent a
    | some winner => if a = winner then S.spent a + charge winner else S.spent a,
  revenue :=
    match scan.winner with
    | none => S.revenue
    | some winner => S.revenue + charge winner,
  decisionsRev := { query := q, winner := scan.winner, spentBefore := S.spent } :: S.decisionsRev,
  candidateTests := S.candidateTests + scan.candidateTests,
  scoreComparisons := S.scoreComparisons + scan.scoreComparisons }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
(feasible : Advertiser → Prop) →
(score : Advertiser → ℝ) →
(a : List Advertiser) →
  List.brecOn a fun x f =>
    (match (motive :=
      (x : List Advertiser) →
        List.below x → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser)
      x with
    | [] => fun x => { winner := none, candidateTests := 0, scoreComparisons := 0 }
    | a :: as => fun x =>
      let tail := x.1;
      if ha : feasible a then
        match tail.winner with
        | none =>
          { winner := some a, candidateTests := tail.candidateTests + 1,
            scoreComparisons := tail.scoreComparisons }
        | some b =>
          if score b ≤ score a then
            { winner := some a, candidateTests := tail.candidateTests + 1,
              scoreComparisons := tail.scoreComparisons + 1 }
          else
            { winner := some b, candidateTests := tail.candidateTests + 1,
              scoreComparisons := tail.scoreComparisons + 1 }
      else
        { winner := tail.winner, candidateTests := tail.candidateTests + 1,
          scoreComparisons := tail.scoreComparisons })
    f
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
  {Query : Type u_2} →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →  
        (q : Query) → (a : Advertiser) → I.bid a q * (1 - Real.exp (S.spent a / I.budget a - 1))
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) (a : Advertiser),
  S.spent a + I.bid a q ≤ I.budget a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [inst : Fintype Advertiser] →  
    [DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →  
          (q : Query) →  
            EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan  
              (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)  
              (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceBalanceScore I S q) Finset.univ.toList
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceStep

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →  
          (q : Query) →  
            EconCSLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision I S q  
              (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceScan I S q)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
(hk : 0 < k) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(q : Query) →
(a : Advertiser) →
I.bid a q *
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteTradeoff k
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a))
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : Fintype Advertiser] →
[DecidableEq Advertiser] →
(k : ℕ) →
(hk : 0 < k) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(q : Query) →
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q)
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceDiscreteScore k hk I S q)
Finset.univ.toList
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : Fintype Advertiser] →
[inst_1 : DecidableEq Advertiser] →
(k : ℕ) →
(hk : 0 < k) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(q : Query) →
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision I S q
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk I S q)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow (N : \mathbb{R}) \rightarrow (\text{stage} : \mathbb{N}) \rightarrow N * (1 - 1 / \uparrow(m + 1)) ^ \text{stage}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow 1 / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow (N : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass } m \ N \ m$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow (N : \mathbb{R}) \rightarrow (i : \text{Fin } m) \rightarrow \text{EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate } N \ i$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(m : ℕ) →
(N : ℝ) →
  ∑ i,
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass m N i *
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction m i +
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass m N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightQueryMassAtSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m : \mathbb{N}) \rightarrow$
 $(N : \mathbb{R}) \rightarrow (stage : \mathbb{N}) \rightarrow EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass\ m\ N\ stage / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(m \text{ stage} : \mathbb{N}) \rightarrow \uparrow \text{stage} / \uparrow (m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_active_mass_drop_realizes_type

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m : ℕ) (N : ℝ) (i : Fin m),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass m N ↑i -
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass m N (↑i + 1) =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass m N i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_budget_feasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m stage : ℕ),
  stage ≤ m →
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab m stage +
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder m ≤
        1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_revenue_eq_N_sub_lpValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m : ℕ) (N : ℝ),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m N =
    N - EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue m N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_slab_work_conservation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m : ℕ) (N : ℝ) (stage : ℕ),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass m N stage *
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder m =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightQueryMassAtSlab m N stage
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_state_update

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m stage : ℕ),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab m (stage + 1) =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab m stage +
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder m
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTight_type_mass_partition

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (m : ℕ) (N : ℝ),
  ∑ i, EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass m N i +
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass m N =
  N
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate_row_tight

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {m : ℕ} (N : ℝ) (i : Fin m),
  EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow (EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate N)
    i =
  EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidExecution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(m : ℕ) →
(N : ℝ) →
{ typeMass := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightTypeMass m N,
  activeMassBeforeSlab := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightActiveMass m N,
  spentBeforeSlab := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightSpentBeforeSlab m,
  allocationPerActiveBidder := fun x =>
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightAllocationPerActiveBidder m,
  queryMassAtSlab := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightQueryMassAtSlab m N,
  fullMass := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFullMass m N,
  revenue := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidRevenue m N,
  typeMass_eq_candidate := "", active_mass_drop_eq_typeMass := "", equal_least_spend_state := "",
  equal_allocation_increment := "", query_mass_formula := "", state_update := "", budget_feasible := "",
  slab_work_conservation := "", type_mass_partition := "", lp_rows_realized_tight := "", revenue_from_final_types := "",
  revenue_from_slab_allocations := "", revenue_eq_N_sub_lpValue := "" }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan._f

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
(feasible : Advertiser → Prop) →
(score : Advertiser → ℝ) →
(x : List Advertiser) →
(f : List.below x) →
  (match (motive :=
    (x : List Advertiser) →
      List.below x → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.FiniteMaxScanResult Advertiser)
    x with
  | [] => fun x => { winner := none, candidateTests := 0, scoreComparisons := 0 }
  | a :: as => fun x =>
    let tail := x.1;
    if ha : feasible a then
      match tail.winner with
      | none =>
        { winner := some a, candidateTests := tail.candidateTests + 1,
          scoreComparisons := tail.scoreComparisons }
      | some b =>
        if score b ≤ score a then
          { winner := some a, candidateTests := tail.candidateTests + 1,
            scoreComparisons := tail.scoreComparisons + 1 }
        else
          { winner := some b, candidateTests := tail.candidateTests + 1,
            scoreComparisons := tail.scoreComparisons + 1 }
    else
      { winner := tail.winner, candidateTests := tail.candidateTests + 1,
        scoreComparisons := tail.scoreComparisons })
  f
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N \ r : \mathbb{N}) \rightarrow \sum \text{round} \in \text{Finset.range } r, 1 / \uparrow(N - \text{round})$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N \ r : \mathbb{N}) \rightarrow \min 1 \text{ (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidHarmonicBefore } N \ r)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N r : ℕ) →  
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N (r + 1) -  
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N r
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (round bidder : Fin N) →  
    if ↑round ≤ ↑bidder then EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceRoundIncrement N ↑round else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_nonneg

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\forall (N : \mathbb{N}) \text{ (round bidder : Fin } N),$
 $0 \leq \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation } N \text{ round bidder}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation_zero_of_ineligible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$\forall (N : \mathbb{N}) (round\ bidder : Fin\ N),$
 $\neg round \leq \uparrow bidder \rightarrow EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation\ N\ round\ bidder = 0$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow (\text{bidder} : \text{Fin } N) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore } N \ (\uparrow \text{bidder} + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (round bidder : Fin N) →  
    if ↑round ≤ ↑bidder then EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N ↑round  
    else EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidCappedSpendBefore N (↑bidder + 1)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_budget_feasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round bidder : Fin N),  
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder +  
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder ≤  
    1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_eligible_equal_least_spend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round bidder bidder' : Fin N),
  ↑round ≤ ↑bidder →
  ↑round ≤ ↑bidder' →
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder =
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder'
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_finalSpend_eq_sum_allocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (bidder : Fin N),  
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend N bidder =  
    ∑ round, EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_allocation_conservation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder ≤
  1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_round_full_or_all_eligible_saturated

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round : Fin N),
  ∑ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder =
      1
  ∀ bidder ∈ EconCSLib.Online.theorem9EligibleBidders N round,
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder +
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder =
      1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_spentBefore_eq_prior_allocations

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round bidder : Fin N),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder =
    ∑ previous,
      if ↑previous < ↑round then
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N previous bidder
      else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalance_state_update

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (N : ℕ) (round next bidder : Fin N),
  rnext = rround + 1 →
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N next bidder =
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N round bidder +
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N round bidder
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
  {Query : Type u_2} →  
    { spent := fun x => 0, revenue := 0, decisionsRev := [], candidateTests := 0, scoreComparisons := 0 }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →
{ allocation := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceAllocation N,
  spentBefore := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceSpentBefore N,
  finalSpend := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.fluidBalanceFinalSpend N, zero_of_ineligible := "",
  eligible_equal_least_spend := "", allocation_nonneg := "", budget_feasible := "", spentBefore_eq_prefix := "",
  state_update := "", round_conservation := "", work_conserving_or_saturated := "", finalSpend_eq_sum := "" }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
(history : List Query) →  
(A : Fin history.length → Option Advertiser) →  
(t : Fin history.length) →  
  match A t with  
  | none => 0  
  | some a => I.bid a (history.get t)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
    (history : List Query) →  
      (A : Fin history.length → Option Advertiser) →  
        ∑ t, EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt I history A t
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
[inst : DecidableEq Advertiser] →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(history : List Query) →
(A : Fin history.length → Option Advertiser) →
(a : Advertiser) →
  ∑ t,
  match A t with
  | none => 0
  | some owner => if owner = a then I.bid a (history.get t) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (m : ℕ) →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      { budget := fun r => I.budget r.1 / ↑m, bid := fun r q => I.bid r.1 q }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  {m : ℕ} →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (rankWeight : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →  
        (r : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) →  
          (q : Query) → I.bid r.1 q * rankWeight r
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
  [inst : Fintype Advertiser] →
  [DecidableEq Advertiser] →
  (m : ℕ) →
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
    (rankWeight : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
    (S :
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query) →
    (q : Query) →
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.finiteMaxScan
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m I) S q)
      (fun r => EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore I rankWeight r q)
    Finset.univ.toList
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : Fintype Advertiser] →
[inst_1 : DecidableEq Advertiser] →
(m : ℕ) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(rankWeight : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →
(S :
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState
  (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) Query) →
(q : Query) →
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.applyScanDecision
  (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m I) S q
  (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScan m I rankWeight S q)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
(step :
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(a : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(a_1 : List Query) →
List.brecOn (motive := fun x =>
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
a_1
(fun x f x_1 =>
  (match (motive :=
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
    (x : List Query) →
    List.below (motive := fun x =>
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
      x →
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
    x_1, x with
  | S, [] => fun x => S
  | S, q :: qs => fun x => x.1 (step S q))
  f)
a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runBalance0ccurrences

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [inst : Fintype Advertiser] →  
  [inst_1 : DecidableEq Advertiser] →  
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
  (history : List Query) →  
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.run0ccurrencesFrom  
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.balanceStep I)  
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initial0ccurrenceState history
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : Fintype Advertiser] →
[inst_1 : DecidableEq Advertiser] →
(k : ℕ) →
(hk : 0 < k) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(history : List Query) →
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep k hk I)
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom._f

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
(step :
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
  Query → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
(x : List Query) →
(f :
  List.below (motive := fun x =>
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
  x) →
(x_1 : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) →
  (match (motive :=
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
    (x : List Query) →
      List.below (motive := fun x =>
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query →
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
        x →
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query)
      x_1, x with
    | S, [] => fun x => S
    | S, q :: qs => fun x => x.1 (step S q))
  f
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runReplicatedRankingProposal

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
  {Query : Type u_4} →  
    [inst : Fintype Advertiser] →  
      [inst_1 : DecidableEq Advertiser] →  
        (m : ℕ) →  
          (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
            (rankWeight : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ) →  
              (history : List Query) →  
                EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom  
                  (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep m I rankWeight)  
                    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runWeightedBidProposal

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (weight : Advertiser → ℝ) →  
          (history : List Query) →  
            EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runBalanceOccurrences  
              (EconCSLib.Online.AdWordsInstance.withAdvertiserWeights I weight) history
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{k : \mathbb{N}\} \rightarrow (N : \mathbb{R}) \rightarrow (x : \text{Fin } k \rightarrow \mathbb{R}) \rightarrow (i : \text{Fin } k) \rightarrow N / \uparrow k - (\sum j \in \text{Finset.Iio } i, x j) / \uparrow k$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
[inst : Fintype Advertiser] →  
  (k : ℕ) →  
    (finalSpentFraction : Advertiser → ℝ) →  
      (i : Fin k) →  
        ∑ a, if EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k (finalSpentFraction a) i then 1 else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
    (d : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) →  
      match d.winner with  
      | none => 0  
      | some a => I.bid a d.query
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (k : ℕ) →  
    (hk : 0 < k) →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (d : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query) →  
        Option.map  
          (fun a =>  
            EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (d.spentBefore a) (I.budget a))  
          d.winner
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
(k : ℕ) →
(hk : 0 < k) →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(decisions : List (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query)) →
(i : Fin k) →
(List.map
  (fun d =>
    if EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk I d = some i then
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue I d
    else 0)
  decisions).sum
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5Delta

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{k : \mathbb{N}\} \rightarrow (\alpha \text{ beta} : \text{Fin } k \rightarrow \mathbb{R}) \rightarrow (i : \text{Fin } k) \rightarrow \sum j \in \text{Finset.Iic } i, (\alpha j - \text{beta } j)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{k : ℕ} →  
(N : ℝ) →  
  (alpha : Fin k → ℝ) →  
    (a : Fin k) → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend N alpha a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section50ptQueryRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
    (history : List Query) →  
      (opt : Fin history.length → Option Advertiser) →  
        (t : Fin history.length) → EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenueAt I history opt t
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section50ptTypeFiberRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
{k : ℕ} →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(history : List Query) →
(opt : Fin history.length → Option Advertiser) →
(finalType : Advertiser → Fin k) →
(i : Fin k) →
  ∑ t,
  match opt t with
  | none => 0
  | some a => if finalType a = i then I.bid a (history.get t) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QueryType

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
  {k : ℕ} → (optOwner : Option Advertiser) → (finalType : Advertiser → Fin k) → Option.map finalType optOwner
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (k : ℕ) →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (i : Fin k) → (a : Advertiser) → I.budget a * ↑↑i / ↑k
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (k : ℕ) →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (i : Fin k) → (a : Advertiser) → I.budget a * ↑(↑i + 1) / ↑k
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  (k : ℕ) →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (finalSpend : Advertiser → ℝ) →  
        (i : Fin k) →  
          (a : Advertiser) →  
            min (finalSpend a) (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper k I i a) -  
              min (finalSpend a) (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower k I i a)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_3} →  
{Query : Type u_4} →  
  [inst : Fintype Advertiser] →  
    (k : ℕ) →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (finalSpend : Advertiser → ℝ) →  
          (i : Fin k) →  
            ∑ a, EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend k I finalSpend i a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6CurrentType

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(k : \mathbb{N}) \rightarrow$
 $(hk : 0 < k) \rightarrow (\text{spent budget} : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab } k \text{ } hk \text{ } \text{spent budget}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section60ptRevenueByFinalType

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{k : ℕ} →
{Advertiser : Type u_3} →
{Query : Type u_4} →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(history : List Query) →
(opt : Fin history.length → Option Advertiser) →
(finalType : Advertiser → Fin k) →
(i : Fin k) →
  ∑ t,
  match opt t with
  | none => 0
  | some a => if finalType a = i then I.bid a (history.get t) else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section60ptRevenueTotalByType

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{k : ℕ} →
{Advertiser : Type u_3} →
{Query : Type u_4} →
[inst : Fintype (Fin k)] →
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
(history : List Query) →
(opt : Fin history.length → Option Advertiser) →
(finalType : Advertiser → Fin k) →
  ∑ i,
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section60ptRevenueByFinalType I history opt finalType i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(N : ℕ) →
 ∑ bidder, (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.nestedSuffixFluidBalanceExecution N).finalSpend bidder

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm N) →  
    (round bidder : Fin N) →  
      EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>  
        algorithm.prefixAllocation (EconCSLib.Online.theorem9ObservedPrefix N permutation round) round  
          (permutation bidder)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardInstance

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N m : ℕ) →  
  (epsilon : ℝ) →  
    (permutation : Equiv.Perm (Fin N)) →  
      { budget := fun x => 1,  
        bid := fun bidder q =>  
          if bidder ∈ EconCSLib.Online.theorem9ActualEligibleBidders N permutation q.1 then epsilon else 0 }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardOfflineAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N\ m : \mathbb{N}) \rightarrow (\text{permutation} : \text{Equiv.Perm } (\text{Fin } N)) \rightarrow (a : \text{Fin } N \times \text{Fin } m) \rightarrow \text{some } (\text{permutation } a.1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →
{
  normalizedRevenue := fun algorithm permutation =>
    (∑ bidder,
      min 1
        (∑ round,
          algorithm.prefixAllocation (EconCSLib.Online.theorem9ObservedPrefix N permutation round) round
            (permutation bidder))) /
    ↑N,
  prefixAllocation := fun algorithm obs round bidder => algorithm.prefixAllocation obs round bidder,
  normalizedRevenue_le_cappedPrefixAllocationSpend := "", prefixAllocation_zero_of_not_visible := "",
  prefixAllocation_sum_le_one := "", revenueBound_le_ratio := "" }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →
  (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralFiniteCertificate
    N).toObservedPrefixAllocationRevenueCertificate.toRelabelSymmetricPointwiseAllocationRevenueCertificate.toSymmetric
    ↪ PointwiseAllocationRevenueCertificate.toPointwiseAllocationRevenueCertificate.toRoundAllocationRevenueCertificate
    ↪ te
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9RoundBid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (epsilon : ℝ) →  
    (permutation : Equiv.Perm (Fin N)) →  
      (round bidder : Fin N) →  
        if permutation bidder ∈ EconCSLib.Online.theorem9ActualEligibleBidders N permutation round then epsilon else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperBalanceScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Query] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) →  
          (a : Advertiser) → (q : Query) → EconCSLib.Online.MSVV07PaperFacing.Proof.paperBalanceScore I A a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperCanAssign

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) (q : Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperCanAssign I A q a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u_1} {Query : Type u_2} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperFeasible I A
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperIsBalanceChoice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Advertiser : Type u 1} {Query : Type u 2} [inst : Fintype Advertiser] [inst_1 : Fintype Query]
  [inst_2 : DecidableEq Advertiser] (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) (q : Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.Proof.paperIsBalanceChoice I A q a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

EconCSLib.Online.MSVV07PaperFacing.Proof.paperMsvvRatio

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Query] →  
    (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
      (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) →  
        EconCSLib.Online.MSVV07PaperFacing.Proof.paperRevenue I A
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperSmallBids

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{ \text{Advertiser} : \text{Type } u_1 \} \{ \text{Query} : \text{Type } u_2 \} (I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query})$
 $(\text{epsilon} : \mathbb{R}), \text{EconCSLib.Online.MSVV07PaperFacing.Proof.paperSmallBids } I \text{ epsilon}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.paperSpend

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Query] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) →  
          (a : Advertiser) → EconCSLib.Online.MSVV07PaperFacing.Proof.paperSpend I A a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.paperTradeoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(s : \mathbb{R}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.Proof.paperTradeoff } s$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →
{Query : Type u_2} →
  [inst : Fintype Advertiser] →
    [inst_1 : DecidableEq Advertiser] →
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →
        (alive : Advertiser → Query → Prop) →
          [inst_2 : (a : Advertiser) → (q : Query) → Decidable (alive a q)] →
            (a : Advertiser) →
              (q : Query) → EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I alive a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Advertiser : Type u_1} →  
{Query : Type u_2} →  
  [inst : Fintype Advertiser] →  
    [inst_1 : DecidableEq Advertiser] →  
      (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) →  
        (a : Advertiser) → (q : Query) → EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.theorem9IntegralPrefixAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9IntegralPrefixAlgorithm } N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07PaperFacing.theorem9CappedNormalizedRevenue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(N : ℕ) →  
  (algorithm : EconCSLib.Online.MSVV07PaperFacing.theorem9IntegralPrefixAlgorithm N) →  
    (permutation : Equiv.Perm (Fin N)) →  
      EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9CappedNormalizedRevenue N algorithm permutation
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.theorem9HardDistribution

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9HardDistribution } N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07PaperFacing.theorem9RandomizedOnlineAlgorithm

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(N : \mathbb{N}) \rightarrow \text{EconCSLib.Online.MSVV07PaperFacing.Proof.theorem9RandomizedOnlineAlgorithm } N$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} → (i j : Fin m) → if ↑j ≤ ↑i then EconCSLib.Online.MSVV07SourceLemmas.paperRouteCoeff i j else 0
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i : \text{Fin } m) \rightarrow \uparrow(m - \uparrow i) / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type u_1 → Type u_2 → Type (max u_1 u_2)

Exact Lean library declaration

```
structure StandardMaxLP (ι κ : Type*) where
  A : κ → ι → ℝ
  b : κ → ℝ
  c : ι → ℝ
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

{ι : Type u_1} → {κ : Type u_2} → (κ → ι → ℝ) → (κ → ℝ) → (ι → ℝ) → EconCSLib.Optimization.StandardMaxLP ι κ

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(m : ℕ) →
(N : ℝ) →
{ A := EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff,
  b := EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N,
  c := EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff }
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDelta

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
(alpha beta : Fin m → ℝ) →  
(i : Fin m) → ∑ j ∈ EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix i, (alpha j - beta j)
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDeltaCoeff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i\ j : \text{Fin } m) \rightarrow (\uparrow\uparrow i - \uparrow\uparrow j) / \uparrow(m + 1)$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i : \text{Fin } m) \rightarrow 1 / \uparrow(m + 1) * (1 - 1 / \uparrow(m + 1)) ^ (m - (\uparrow i + 1))$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteSuffix

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (i : \text{Fin } m) \rightarrow \{j \mid \uparrow i \leq \uparrow j\}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualRow

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
(y : Fin m → ℝ) →  
(i : Fin m) →  
  ∑ j ∈ EconCSLib.Online.MSVV07SourceLemmas.paperRouteSuffix i,  
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteCoeff j i * y j
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {m : ℕ} (y : Fin m → ℝ),
  (∀ (i : Fin m),
    EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff i ≤
      EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualRow y i) ∧
  ∀ (i : Fin m), 0 ≤ y i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (N : \mathbb{R}) \rightarrow (y : \text{Fin } m \rightarrow \mathbb{R}) \rightarrow \sum i, \text{EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs } N \ i * y \ i$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {m : ℕ} (N : ℝ) (x : Fin m → ℝ),  
  (∀ (i : Fin m),  
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow x i ≤ EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i) ∧  
    ∀ (i : Fin m), 0 ≤ x i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (x : \text{Fin } m \rightarrow \mathbb{R}) \rightarrow \sum i, \text{EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjectiveCoeff } i * x \ i$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (y : \text{Fin } m \rightarrow \mathbb{R}) \rightarrow (i : \text{Fin } m) \rightarrow \sum j \in \text{EconCSLib.Online.MSVV07SourceLemmas.paperRouteSuffix } i, y \ j$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →  
(a : Fin m) →  
  EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual  
  EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{m : \mathbb{N}\} \rightarrow (c \ l : \text{Fin } m \rightarrow \mathbb{R}) \rightarrow \{ A := \text{EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff}, b := l, c := c \}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.IsMaximizerOn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} \text{ (feasible : } \alpha \rightarrow \text{Prop) (objective : } \alpha \rightarrow \mathbb{R}) (x : \alpha),$
 $\text{feasible } x \wedge \forall (y : \alpha), \text{feasible } y \rightarrow \text{objective } y \leq \text{objective } x$

Exact Lean library declaration

```
def IsMaximizerOn {α : Type*} (feasible : α → Prop)
  (objective : α → ℝ) (x : α) : Prop :=
  feasible x ∧ ∀ y, feasible y → objective y ≤ objective x
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.IsMinimizerOn

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} \text{ (feasible : } \alpha \rightarrow \text{Prop) (objective : } \alpha \rightarrow \mathbb{R}) (x : \alpha),$
 $\text{feasible } x \wedge \forall (y : \alpha), \text{feasible } y \rightarrow \text{objective } x \leq \text{objective } y$

Exact Lean library declaration

```
def IsMinimizerOn {α : Type*} (feasible : α → Prop)
  (objective : α → ℝ) (x : α) : Prop :=
  feasible x ∧ ∀ y, feasible y → objective x ≤ objective y
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.A

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → {κ : Type u_2} → (self : EconCSLib.Optimization.StandardMaxLP ι κ) → (a : κ) → (a_1 : ι) → self.1 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.c

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → {κ : Type u_2} → (self : EconCSLib.Optimization.StandardMaxLP ι κ) → (a : ι) → self.3 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.DualFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {ι : Type u_1} {κ : Type u_2} [inst : Fintype κ] (P : EconCSLib.Optimization.StandardMaxLP ι κ) (y : κ → ℝ),  
  (∀ (k : κ), 0 ≤ y k) ∧ ∀ (i : ι), P.c i ≤ ∑ k, P.A k i * y k
```

Exact Lean library declaration

```
def DualFeasible (P : StandardMaxLP ι κ) (y : κ → ℝ) : Prop :=  
  (∀ k, 0 ≤ y k) ∧ ∀ i, P.c i ≤ ∑ k : κ, P.A k i * y k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.b

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → {κ : Type u_2} → (self : EconCSLib.Optimization.StandardMaxLP ι κ) → (a : κ) → self.2 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.PrimalFeasible

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {ι : Type u_1} {κ : Type u_2} [inst : Fintype ι] (P : EconCSLib.Optimization.StandardMaxLP ι κ) (x : ι → ℝ),  
  (∀ (i : ι), 0 ≤ x i) ∧ ∀ (k : κ), ∑ i, P.A k i * x i ≤ P.b k
```

Exact Lean library declaration

```
def PrimalFeasible (P : StandardMaxLP ι κ) (x : ι → ℝ) : Prop :=  
  (∀ i, 0 ≤ x i) ∧ ∀ k, (∑ i : ι, P.A k i * x i) ≤ P.b k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.dualObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} →
{κ : Type u_2} → [inst : Fintype κ] → (P : EconCSLib.Optimization.StandardMaxLP ι κ) → (y : κ → ℝ) → ∑ k, y k * P.b k
```

Exact Lean library declaration

```
noncomputable def dualObjective (P : StandardMaxLP ι κ)
  (y : κ → ℝ) : ℝ :=
  ∑ k : κ, y k * P.b k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Optimization.StandardMaxLP.primalObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} →  
{κ : Type u_2} → [inst : Fintype ι] → (P : EconCSLib.Optimization.StandardMaxLP ι κ) → (x : ι → ℝ) → ∑ i, P.c i * x i
```

Exact Lean library declaration

```
noncomputable def primalObjective (P : StandardMaxLP ι κ)  
  (x : ι → ℝ) : ℝ :=  
  ∑ i : ι, P.c i * x i
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Sequence.SeqTendsTo

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (x : \mathbb{N} \rightarrow \mathbb{R}) (L \ \delta : \mathbb{R}), \ 0 < \delta \rightarrow \exists N, \ \forall (k : \mathbb{N}), \ N \leq k \rightarrow |x \ k - L| \leq \delta$

Exact Lean library declaration

```
def SeqTendsTo (x : ℕ → ℝ) (L : ℝ) : Prop :=
  ∀ δ : ℝ, 0 < δ → ∃ N : ℕ, ∀ k : ℕ, N ≤ k → |x k - L| ≤ δ
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 198-205

Verbatim source input

The adwords problem is the following: There are N bidders, each with a specified daily budget b_i . Q is a set of query words. Each bidder i specifies a bid c_{iq} for query word $q \in Q$. A sequence $q_1 q_2 \dots q_M$ of query words $q_j \in Q$ arrive online during the day, and each query q_j must be assigned to some bidder i (for a revenue of c_{iq_j}). The objective is to maximize the total revenue at the end of the day while respecting the daily budgets of the bidders. Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query),
  EconCSLib.Online.MSVV07PaperFacing.paperFeasible I A ↔
    ∀ (a : Advertiser), EconCSLib.Online.MSVV07PaperFacing.paperSpend I A a ≤ I.budget a
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperFeasible_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 198-208

Verbatim source input

The adwords problem is the following: There are N bidders, each with a specified daily budget b_i . Q is a set of query words. Each bidder i specifies a bid c_{iq} for query word $q \in Q$. A sequence $q_1 q_2 \dots q_M$ of query words $q_j \in Q$ arrive online during the day, and each query q_j must be assigned to some bidder i (for a revenue of c_{ij}). The objective is to maximize the total revenue at the end of the day while respecting the daily budgets of the bidders. Throughout this paper we will make the assumption that each bid is small compared to the corresponding budget, i.e., $\max_j c_{ij}$ is small compared to b_i , for all i . For the applications of this problem mentioned in the Introduction, this is a reasonable assumption. An online algorithm is said to be α -competitive if for every instance, the ratio of the revenue of the online algorithm to the revenue of the best off-line algorithm is at least α . While presenting the algorithm and the proofs, we will make the simplifying assumptions that

Expanded PaperInterface specification

$\forall \{ \text{Instance} : \text{Type } u \}$ (onlineRevenue offlineRevenue : Instance $\rightarrow \mathbb{R}$) ($\alpha : \mathbb{R}$),
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsAlphaCompetitive onlineRevenue offlineRevenue $\alpha \leftrightarrow$
 $\forall (I : \text{Instance}), \alpha \leq \text{onlineRevenue } I / \text{offlineRevenue } I$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section2_competitive_ratio_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 225-239

Verbatim source input

Our algorithm rectifies this situation by taking into consideration not only the bids but also the unspent budget of each bidder. For the analysis it is convenient to discretize the budgets as follows: we pick a large integer k , and discretize the budget of each bidder into k equal

[form-feed in source extraction]
parts (called slabs) numbered 1 through k . Each bidder spends money in slab j before moving to slab $j + 1$.
Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}_+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.
Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

$\forall (k : \mathbb{N}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff $k \ i = 1 - \text{Real.exp } (-(1 - \uparrow(i + 1) / \uparrow k))$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 232-236

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.

Expanded PaperInterface specification

$\forall (k : \mathbb{N}), \text{Antitone } (\text{EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperDiscreteTradeoff } k)$

Lean proof endpoint (built separately)

`EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_monotonicity`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 232-239

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}^+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.
Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Query] [inst_1 : DecidableEq Advertiser]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) (a : Advertiser) (q : Query),
EconCSLib.Online.MSVV07PaperFacing.paperBalanceScore I A a q =
  I.bid a q *
  EconCSLib.Online.MSVV07PaperFacing.paperTradeoff
    (EconCSLib.Online.MSVV07PaperFacing.paperSpend I A a / I.budget a)
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperBalanceScore_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 237-239

Verbatim source input

Discrete Version of the Algorithm

When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Fintype Query]
  [inst_2 : DecidableEq Advertiser] (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (A : EconCSLib.Online.MSVV07PaperFacing.PaperAssignment Advertiser Query) (q : Query) (a : Advertiser),
  EconCSLib.Online.MSVV07PaperFacing.paperIsBalanceChoice I A q a ↔
    EconCSLib.Online.MSVV07PaperFacing.paperCanAssign I A q a ∧
      ∀ (b : Advertiser),
        EconCSLib.Online.MSVV07PaperFacing.paperCanAssign I A q b →
          EconCSLib.Online.MSVV07PaperFacing.paperBalanceScore I A b q ≤
            EconCSLib.Online.MSVV07PaperFacing.paperBalanceScore I A a q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.paperIsBalanceChoice_iff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 214-225

Verbatim source input

A Discretized Version of the Algorithm

Let us first consider a greedy algorithm that maximizes revenue accrued at each step. It is easy to see that this algorithm achieves a competitive ratio of 21 (see, e.g., [LLN01]); moreover, this is tight as shown by the following example with only two bidders and two query words: Suppose both bidders have unit budget. The two bidders bid c and $c + \text{[U+000F control character]}$ respectively on query word q , and they bid 0 and c on query word $q \ 0$. The query sequence consists of a number of occurrences of q followed by a number of occurrences of $q \ 0$. The query words q are awarded to bidder 2, and are just enough in number to exhaust his budget. When query words $q \ 0$ arrive, bidder 2's budget is exhausted and bidder 1 is not interested in this query word, and they accrue no further revenue. Our algorithm rectifies this situation by taking into consideration not only the bids but also

Expanded PaperInterface specification

```
∀ {c : ℝ},
  0 < c →
  Filter.Tendsto
    (fun epsilon => EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3GreedyTwoWordRatio c epsilon) (nhds 0)
    (nhds (1 / 2))
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_greedy_two_word_fluid_construction

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 225-233

Verbatim source input

Our algorithm rectifies this situation by taking into consideration not only the bids but also the unspent budget of each bidder. For the analysis it is convenient to discretize the budgets as follows: we pick a large integer k , and discretize the budget of each bidder into k equal

[form-feed in source extraction]
parts (called slabs) numbered 1 through k . Each bidder spends money in slab j before moving to slab $j + 1$.
Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .

Expanded PaperInterface specification

$\forall (k : \mathbb{N}) (hk : 0 < k) (\text{spentFraction} : \mathbb{R}),$
 $\uparrow(\text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeSlab } k \text{ } hk \text{ } \text{spentFraction}) =$
 $\min [\max 0 (\uparrow k * \text{spentFraction})]_+ (k - 1)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_slab_and_current_slab_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 232-239

Verbatim source input

Definition: At any time during the run of the algorithm, we will denote by $\text{slab}(i)$ the currently active slab for bidder i .
Let $\psi_k : [1 \dots k] \rightarrow \mathbb{R}_+$ be the following (monotonically decreasing) function:
 $\psi_k(i) = 1 - e^{-(1-i/k)}$
Note that $\psi_k \rightarrow \psi$ as $k \rightarrow \infty$.
Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.

Expanded PaperInterface specification

$\forall (\text{slab} : \mathbb{N} \rightarrow \mathbb{N}) (s : \mathbb{R}),$
Filter.Tendsto (fun k => ↑(slab k + 1) / ↑(k + 1)) Filter.atTop (nhds s) →
EconCSLib.Sequence.SeqTendsTo (fun k => 1 - Real.exp (-(1 - ↑(slab k + 1) / ↑(k + 1)))) (1 - Real.exp (-(1 - s)))

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_tradeoff_convergence

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 237-241

Verbatim source input

Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.
Note that in the special case when all the bids are equal, our algorithm works in the same way as the BALANCE algorithm of [KP00], for any monotonically decreasing tradeoff function.

Expanded PaperInterface specification

```

V {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (k : N)
(hk : 0 < k) (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query),
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk I history =
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep k hk I)
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_discrete_balance_algorithm

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 237-245

Verbatim source input

Discrete Version of the Algorithm
When a new query arrives, let the bid of bidder i be $c(i)$. Allocate the query to the bidder i who maximizes $c(i) \times \psi_k(\text{slab}(i))$.
Note that in the special case when all the bids are equal, our algorithm works in the same way as the BALANCE algorithm of [KP00], for any monotonically decreasing tradeoff function.

4

Analyzing BALANCE using a Factor-Revealing LP

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (k : ℕ)
(hk : 0 < k) (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(S : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceState Advertiser Query) (q : Query) (a : Advertiser)
{commonBid : ℝ},
(∀ (b : Advertiser), I.bid b q = commonBid) →
0 < commonBid →
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceScan k hk I S q).winner = some a →
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q a ∧
  ∀ (b : Advertiser),
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceCanAssign I S q b →
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent a) (I.budget a) ≤
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (S.spent b) (I.budget b)

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section3_equal_bid_discrete_msvv_is_balance

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 833-841

Verbatim source input

6. A winning bidder pays only an amount equal to the next highest bid.
1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
13

[form-feed in source extraction]
The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval

Expanded PaperInterface specification

$\forall (k : \mathbb{N}) (hk : 0 < k) (\text{spent budget} : \mathbb{R}),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6CurrentType k hk $\text{spent budget} =$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk spent budget

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_current_type_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 833-842

Verbatim source input

6. A winning bidder pays only an amount equal to the next highest bid.
1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
13

[form-feed in source extraction]
The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval P_j

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (k : ℕ)
  (hk : 0 < k) (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk I history =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.discreteBalanceStep k hk I)
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_generalized_current_type_balance_algorithm

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 842-850

Verbatim source input

P_j
 i
[$i-1$
 j β_i . Let α_i be the
 k B_j , k B_j) of his budget. Let $\beta_i =$
 P amount of money that the optimal
allocation gets from the bins of final type i . Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become

Expanded PaperInterface specification

$\forall$ {Advertiser : Type u_5 } {Query : Type u_6 } ($k : \mathbb{N}$)
($I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}$) ($\text{finalSpend} : \text{Advertiser} \rightarrow \mathbb{R}$) ($i : \text{Fin } k$)
($a : \text{Advertiser}$),
 $\text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend } k \text{ } I \text{ } \text{finalSpend } i \text{ } a =$
 $\min (\text{finalSpend } a) (\text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabUpper } k \text{ } I \text{ } i \text{ } a) -$
 $\min (\text{finalSpend } a) (\text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6SlabLower } k \text{ } I \text{ } i \text{ } a)$

Lean proof endpoint (built separately)

`EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_beta_ij_definition`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 842-850

Verbatim source input

P j
i
[i-1
j β_i . Let α_i be the
k Bj , k Bj) of his budget. Let β_i =
P amount of money that the optimal
allocation gets from the bins of final type i. Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become

Expanded PaperInterface specification

$\forall$ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] (k : $\mathbb{N}$)
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (finalSpend : Advertiser $\rightarrow$ $\mathbb{R}$) (i : Fin k),
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend k I finalSpend i =
 $\sum a$, EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6BidderSlabSpend k I finalSpend i a

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_beta_i_aggregate_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 842-852

Verbatim source input

P j
i
[i-1
j β_i . Let α_i be the
k Bj , k Bj) of his budget. Let β_i =
P amount of money that the optimal
allocation gets from the bins of final type i. Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
P
 $\alpha - ij=1 \alpha_j$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} {k : ℕ}
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k) (i : Fin k),
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section60ptRevenueByFinalType I history opt finalType i =
  ∑ t,
    match opt t with
    | none => 0
    | some a => if finalType a = i then I.bid a (history.get t) else 0
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_alpha_i_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 848-853

Verbatim source input

allocation gets from the bins of final type i . Let $\alpha = \sum_i \alpha_i$, be the total amount of money obtained in the optimal allocation.
Now the relations used in the direct proof at the end of Section 5 become
$$\sum_j \alpha_j = \alpha$$
$$\forall i : \beta_i \geq \alpha_i$$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} {k : ℕ}
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
  (opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType I history opt finalType =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceRevenue I history opt
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_alpha_total_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 852-860

Verbatim source input

$$\alpha - \sum_{i=1}^k \beta_i \geq \sum_{i=1}^k \psi(i) \alpha_i \leq \sum_{i=1}^k \psi(i) \beta_i$$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] {k : ℕ}
(hk : 0 < k) (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (history : List Query)
(opt : Fin history.length → Option Advertiser) (finalType : Advertiser → Fin k),
EconCSLib.Online.AdWordsInstance.PositiveBudgets I →
  (∀ (a : Advertiser), EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend I history opt a ≤ I.budget a) →
  (∀ (a : Advertiser),
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k
      ((EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk I history).spent a /
        I.budget a)
    (finalType a)) →
  EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6BetaLowerBounds
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueTotalByType I history opt finalType)
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6OptRevenueByFinalType I history opt finalType)
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section6AggregateSlabSpend k I
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runDiscreteBalanceOccurrences k hk I history).spent)
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_beta_lower_bound_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 852-861

Verbatim source input

$\alpha - \text{ij}=1 \alpha_j$
 $\forall i : \beta_i \geq$
 k
 X
 X
 $\psi(i)\alpha_i \leq$
 $\psi(i)\beta_i$
 i

 i

Expanded PaperInterface specification

$\forall \{m : \text{Type } u_5\} \{ \text{Query} : \text{Type } u_6\} [\text{inst} : \text{Fintype } m] [\text{inst}_1 : \text{Fintype } \text{Query}] (\psi \text{ alpha beta} : m \rightarrow \mathbb{R})$
 $(\text{opt alg} : \text{Query} \rightarrow \mathbb{R}) (\text{queryType querySlab} : \text{Query} \rightarrow m),$
 $\sum q, (\text{opt } q * \psi (\text{queryType } q) - \text{alg } q * \psi (\text{querySlab } q)) \leq 0 \rightarrow$
 $\sum q, \text{opt } q * \psi (\text{queryType } q) = \sum i, \psi i * \text{alpha } i \rightarrow$
 $\sum q, \text{alg } q * \psi (\text{querySlab } q) \leq \sum i, \psi i * \text{beta } i \rightarrow \sum i, \psi i * \text{alpha } i \leq \sum i, \psi i * \text{beta } i$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_weighted_alpha_beta_inequality

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 825-865

Verbatim source input

In this section we show how our algorithm and analysis can be generalized to the following situations:

1. Advertisers have different daily budgets.
 2. The optimal allocation does not exhaust all the money of advertisers
 3. Advertisers enter at different times.
 4. More than one ad can appear with the results of a query. The most general situation is that with each query we are provided a number specifying the maximum number of ads.
 5. A bidder pays only if the user clicks on his ad.
 6. A winning bidder pays only an amount equal to the next highest bid.
- 1, 2, 3: We say that the current type of a bidder at some time during the run of the algorithm is j if he has spent between $(j-1)/k$ and j/k fraction of his budget at that time. The algorithm allocates the next query to the bidder who maximizes the product of his bid and $\psi(\text{current type})$.
- 13

[form-feed in source extraction]

The proof of the competitive ratio changes minimally: Let the budget of bidder j be B_j . For $i = 1, \dots, k$, define β_{ij} to be the amount of money spent by the bidder j from the interval

P_j
 i
[$i-1$
 $j \beta_i$. Let α_i be the
 $k B_j$, $k B_j$) of his budget. Let $\beta_i =$
 P amount of money that the optimal
allocation gets from the bins of final type i . Let $\alpha = i \alpha_i$, be the total amount of money
obtained in the optimal allocation.

Now the relations used in the direct proof at the end of Section 5 become

$\alpha - i j = 1 \alpha_j$

$\forall i : \beta_i \geq$

k

X

X

$\psi(i) \alpha_i \leq$

$\psi(i) \beta_i$

i

i

These two sets of equations suffice to prove that the competitive ratio is at least $1 - 1/e$. We also note that the algorithm and the proof of the competitive ratio remain unchanged even if we allow advertisers to enter the bidding process at any time during the query sequence.

Expanded PaperInterface specification

```

V {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : Fintype Query] [inst_3 : DecidableEq Advertiser] [inst_4 : DecidableEq Query]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query),
EconCSLib.Online.AdWordsInstance.NonnegativeBids I →
  V (hbudget : EconCSLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query),
    history.Nodup →
      EconCSLib.Online.AdWordsInstance.historyFinset history = Finset.univ →
        V {epsilon : ℝ},
          0 ≤ epsilon →
            epsilon ≤ 1 →
              EconCSLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
                EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio *
                  EconCSLib.Online.AdWordsInstance.offlineOptimumValue I ≤
                    EconCSLib.Online.MSVV07PaperFacing.paperRevenue I
                      (EconCSLib.Online.AdWordsInstance.runAssignment I
                        (EconCSLib.Online.AdWordsInstance.balanceChoiceRule I) history) +
                        epsilon * (Real.exp 1 + 1) * ∑ q, EconCSLib.Online.AdWordsInstance.maxBidForQuery I q

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_different_budgets_and_nonexhaustive_optimum_theorem8_finite_explicit_e_j
↔ rror

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 872-879

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi(\text{fraction of money spent})$. Next, let us consider the second possibility. In this case, the offline

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (a : Advertiser) (q : Query),
EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I a q =
  have others := Finset.univ.erase a;
  if h : others.Nonempty then max 0 (others.sup' h fun b => I.bid b q) else 0
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_all_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 872-882

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi(\text{fraction of money spent})$. Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u 5} {Query : Type u 6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (alive : Advertiser → Query → Prop)
[inst_2 : (a : Advertiser) → (q : Query) → Decidable (alive a q)] (a : Advertiser) (q : Query),
EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I alive a q =
  have others := {b ∈ Finset.univ.erase a | alive b q};
  if h : others.Nonempty then max 0 (others.sup' h fun b => I.bid b q) else 0
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_alive_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 877-885

Verbatim source input

It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi(\text{fraction of money spent})$. Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

7

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (a : Advertiser) (q : Query),
  EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_alive I (fun x x_1 => True) a q =
    EconCSLib.Online.MSVV07PaperFacing.Proof.section6_next_highest_bid_all I a q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_alive_keep_alive_reduction

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 872-879

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi(\text{fraction of money spent})$. Next, let us consider the second possibility. In this case, the offline

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : Fintype Query] [inst_3 : DecidableEq Advertiser] [inst_4 : DecidableEq Query]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
(hbudget : EconCSLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query),
history.Nodup →
  EconCSLib.Online.AdWordsInstance.historyFinset history = Finset.univ →
    ∀ {epsilon : ℝ},
      0 ≤ epsilon →
        epsilon ≤ 1 →
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets I →
            EconCSLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
              EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio *
                (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                  (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I)).offlineOptimumValue
                " ≤
            EconCSLib.Online.MSVV07PaperFacing.paperRevenue
              (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I))
              ((EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I)).runAssignment
                (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                  (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I)).balanceChoiceRule
                history) +
              epsilon * (Real.exp 1 + 1) *
                ∑ q,
                  (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                    (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_all I)).maxBidForQuery
                    q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_all_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 872-882

Verbatim source input

6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query. Search engine companies charge a lower amount: the next highest bid. There are different ways of defining “next highest bid”. We can extend our analysis for two of these definitions: the next highest bid is chosen from all bids received at the start of the algorithm or only among alive bidders, i.e. bidders who still have money. It is easy to see that a small modification of our algorithm achieves a competitive ratio of $1-1/e$ for the first possibility: award the query to the bidder that maximizes next highest bid $\times \psi(\text{fraction of money spent})$. Next, let us consider the second possibility. In this case, the offline algorithm will attempt to keep alive bidders simply to charge other bidders higher amounts. If the online algorithm is also allowed this capability, it can also keep all bidders alive all the way to the end and this possibility reduces to the first one.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : Fintype Query] [inst_3 : DecidableEq Advertiser] [inst_4 : DecidableEq Query]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (alive : Advertiser → Query → Prop)
[inst_5 : (a : Advertiser) → (q : Query) → Decidable (alive a q)]
(hbudget : EconCSLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query),
history.Nodup →
  EconCSLib.Online.AdWordsInstance.historyFinset history = Finset.univ →
    ∀ {epsilon : ℝ},
      0 ≤ epsilon →
        epsilon ≤ 1 →
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.EqualUnitBudgets I →
            EconCSLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
              EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio *
                (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                  (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I
                    alive)).offlineOptimumValue
                  "" ≤
                    EconCSLib.Online.MSVV07PaperFacing.paperRevenue
                      (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                        (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I alive))
                      ((EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                        (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I alive)).runAssignment
                        (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                          (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I
                            alive)).balanceChoiceRule
                            history) +
                          epsilon * (Real.exp 1 + 1) *
                          ∑ q,
                            (EconCSLib.Online.AdWordsInstance.withEffectiveBids I
                              (EconCSLib.Online.MSVV07PaperFacing.section6_next_highest_bid_alive I
                                alive)).maxBidForQuery
                                q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_next_highest_bid_alive_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 866-872

Verbatim source input

4: If the arriving query q requires n_q number of advertisements to be placed, then allocate it to the bidders with the top n_q values of the product of bid and $\psi(\text{current type})$. The proof of the competitive ratio remains unchanged.
5: In order to model this situation, we simply set the effective bid of a bidder to be the product of his actual bid and his click-through rate (CTR), which is the probability that a user will click on his ad. We assume that the click-through rate is known to the algorithm in advance indeed several search engines keep a $\rightarrow$ measure of the click-through rates of the bidders.
6: So far we have assumed that a bidder is charged the value of his bid if he is awarded a query.

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \} (I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query})$
 $(\text{ctr} : \text{Advertiser} \rightarrow \text{Query} \rightarrow \mathbb{R}) (a : \text{Advertiser}) (q : \text{Query}),$
 $(\text{EconCSLib.Online.AdWordsInstance.withClickThroughRates } I \text{ ctr}).\text{bid } a \text{ } q = \text{ctr } a \text{ } q * I.\text{bid } a \text{ } q$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_click_through_rates_effective_bid_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 869-871

Verbatim source input

5: In order to model this situation, we simply set the effective bid of a bidder to be the product of his actual bid and his click-through rate (CTR), which is the probability that a user will click on his ad. We assume that the click-through rate is known to the algorithm in advance indeed several search engines keep a ↪ measure of the click-through rates of the bidders.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : Fintype Query] [inst_3 : DecidableEq Advertiser] [inst_4 : DecidableEq Query]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (ctr : Advertiser → Query → ℝ),
(∀ (a : Advertiser) (q : Query), 0 ≤ ctr a q) →
(∀ (a : Advertiser) (q : Query), ctr a q ≤ 1) →
EconCSLib.Online.AdWordsInstance.NonnegativeBids I →
  ∀ (hbudget : EconCSLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query),
    history.Nodup →
      EconCSLib.Online.AdWordsInstance.historyFinset history = Finset.univ →
        ∀ {epsilon : ℝ},
          0 ≤ epsilon →
            epsilon ≤ 1 →
              EconCSLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
                EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio *
                  (EconCSLib.Online.AdWordsInstance.withClickThroughRates I ctr).offlineOptimumValue ... ≤
                    (EconCSLib.Online.AdWordsInstance.withClickThroughRates I ctr).revenue
                      ((EconCSLib.Online.AdWordsInstance.withClickThroughRates I ctr).runAssignment
                        (EconCSLib.Online.AdWordsInstance.withClickThroughRates I ctr).balanceChoiceRule
                          history) +
                        epsilon * (Real.exp 1 + 1) *
                          ∑ q, (EconCSLib.Online.AdWordsInstance.withClickThroughRates I ctr).maxBidForQuery q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_click_through_rates_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 863-865

Verbatim source input

These two sets of equations suffice to prove that the competitive ratio is at least $1 - 1/e$. We also note that the algorithm and the proof of the competitive ratio remain unchanged even if we allow advertisers to enter the bidding process at any time during the query sequence.

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : Nonempty Advertiser]
[inst_2 : Fintype Query] [inst_3 : DecidableEq Advertiser] [inst_4 : DecidableEq Query]
(I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (available : Advertiser → Query → Prop)
[inst_5 : (a : Advertiser) → (q : Query) → Decidable (available a q)],
EconCSLib.Online.AdWordsInstance.NonnegativeBids I →
  ∀ (hbudget : EconCSLib.Online.AdWordsInstance.PositiveBudgets I) (history : List Query),
    history.Nodup →
      EconCSLib.Online.AdWordsInstance.historyFinset history = Finset.univ →
        ∀ {epsilon : ℝ},
          0 ≤ epsilon →
            epsilon ≤ 1 →
              EconCSLib.Online.MSVV07PaperFacing.paperSmallBids I epsilon →
                EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio *
                  ((EconCSLib.Online.AdWordsInstance.withAvailability I available).offlineOptimumValue ... ≤
                    (EconCSLib.Online.AdWordsInstance.withAvailability I available).revenue
                     ((EconCSLib.Online.AdWordsInstance.withAvailability I available).runAssignment
                      (EconCSLib.Online.AdWordsInstance.withAvailability I available).balanceChoiceRule history) +
                     epsilon * (Real.exp 1 + 1) *
                     ∑ q, (EconCSLib.Online.AdWordsInstance.withAvailability I available).maxBidForQuery q
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section6_availability_theorem8_finite_explicit_error

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 949-960

Verbatim source input

offline problem. Though this is NP-complete for large bids, in the realistic case where the bids are small compared to budgets, a $1 - \epsilon$ control character approximation can be obtained by linear programming [BHJ+ 04]. In practice, the query distribution fluctuates over time, varying with time of day, special events, etc. Therefore it is desirable to have a very simple, time efficient online allocation scheme.

Let us start by giving a model for the query sequence that formalizes both their statistical nature as well as their unpredictable fluctuations. In this model, the queries are drawn from an arbitrary fixed distribution for a period of time. The distribution is switched by an adversary a number of times each day. Our goal is to design an online algorithm that achieves a $1 - o(1)$ performance ratio against any fixed probability distribution, while achieving a good worst-case performance ratio when the distribution is switched suddenly (or evolves rapidly).

We believe a simple modification of our algorithm achieves this. Here is a concrete proposal:

Expanded PaperInterface specification

```
∀ {Query : Type u 2} {Time : Type u 3} {Regime : Type u 4}
  (M : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.SwitchingQueryDistribution Time Regime Query) (time : Time),
  M.queryDistributionAt time = M.queryDistribution (M.regimeAt time) ∧
  M = { regimeAt := M.regimeAt, queryDistribution := M.queryDistribution }
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section8_switching_query_distribution_model

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 960-965

Verbatim source input

We believe a simple modification of our algorithm achieves this. Here is a concrete proposal: each bidder is assigned a weight, and his effective bid for a keyword is defined to be the product of the actual bid and his weight. We modify our algorithm to use effective bids in place of bids. The main open question here is whether for any fixed distribution on queries there is always a set of weights such that this algorithm achieves $1 - o(1)$ expected competitive ratio. How do we actually find a good set of weights for the bidders? Here is an online heuristic

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \} (I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query})$
 $(\text{weight} : \text{Advertiser} \rightarrow \mathbb{R}) (a : \text{Advertiser}) (q : \text{Query}),$
 $(\text{EconCSLib.Online.AdWordsInstance.withAdvertiserWeights } I \text{ weight}).\text{bid } a \text{ } q = \text{weight } a * I.\text{bid } a \text{ } q$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section8_weighted_effective_bid_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 960-966

Verbatim source input

We believe a simple modification of our algorithm achieves this. Here is a concrete proposal: each bidder is assigned a weight, and his effective bid for a keyword is defined to be the product of the actual bid and his weight. We modify our algorithm to use effective bids in place of bids. The main open question here is whether for any fixed distribution on queries there is always a set of weights such that this algorithm achieves $1 - o(1)$ expected competitive ratio. How do we actually find a good set of weights for the bidders? Here is an online heuristic might provide a quick way of computing such weights: consider the allocation of queries for

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser]
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query) (weight : Advertiser → ℝ)
  (history : List Query),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runWeightedBidProposal I weight history =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runBalanceOccurrences
      (EconCSLib.Online.AdWordsInstance.withAdvertiserWeights I weight) history
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section8_weighted_bid_algorithm_proposal

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 965-971

Verbatim source input

How do we actually find a good set of weights for the bidders? Here is an online heuristic might provide a quick way of computing such weights: consider the allocation of queries for some window of time under the current weights. Adjust the weight of a bidder upwards if that bidder spends less than his fair share of his his budget during this time window, and downwards if he spends more than his fair share of the budget. Repeat this process after each such window of time.
In an earlier version of this paper [MSVV05], we had presented a second algorithm based on

Expanded PaperInterface specification

```
∀ (weight finalWeight : ℝ) (windows : List (ℝ × ℝ)),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment weight windows finalWeight ↔
    match windows with
    | [] => finalWeight = weight
    | (spent, fairShare) :: remaining =>
      ∃ nextWeight,
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsWeightAdjustment weight spent fairShare nextWeight ∧
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsRepeatedWeightAdjustment nextWeight remaining finalWeight
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section8_online_weight_adjustment_heuristic

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 984-988

Verbatim source input

However, there is yet another way to generalize RANKING for general bids and large budgets. For a suitably large m , we can represent every bidder i with budget B_i by m bidders each with budget B_i / m and the same bid values. Now we can run the previous generalization of RANKING for new bidders. As the queries arrive and get allocated by the algorithm, the representatives of each bidder will run out of budget in the order in which they appear in the

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_5} {Query : Type u_6} [inst : Fintype Advertiser] [inst_1 : DecidableEq Advertiser] (m : ℕ),
  0 < m →
  ∀ (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
    (rankWeight : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m → ℝ)
    (history : List Query),
  (∀ (r : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m),
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m I).budget r = I.budget r.1 / ↑m) ∧
  (∀ (r : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.Representative Advertiser m) (q : Query),
    (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedInstance m I).bid r q = I.bid r.1 q ∧
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingScore I rankWeight r q =
      I.bid r.1 q * rankWeight r) ∧
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runReplicatedRankingProposal m I rankWeight history =
    EconCSLib.Online.MSVV07PaperFacing.SourceRunner.runOccurrencesFrom
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.replicatedRankingStep m I rankWeight)
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.initialOccurrenceState history
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section8_replicated_ranking_algorithm_proposal

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 1176-1213

Verbatim source input

Bin N

Bin $0.5N$

Bin $0.1N$ Bin 1

Figure 2: The bidders are ordered from right to left. The area inside the dark outline is the amount of money generated by the algorithm. The optimum allocation gets an amount equal to the whole rectangle.

We present an example to show a factor strictly less than $1 - 1/e$ for the algorithm which gives a query to a highest bidder, breaking ties by giving it to the bidder with most left-over money. This example has only three values for the bids - 0 , a or $2a$, for some small $a > 0$. Thus, in the case of arbitrary bids, the strategy of bucketing close enough bids (say within a factor of 2) together, and running such an algorithm does not work.

There are N bidders numbered $1, \dots, N$, each with budget 1. We get the following query sequence and bidding pattern. Each bid is either 0 , a or $2a$. Let $m = 1/a$. We will take $a \rightarrow 0$. The queries arrive in N rounds. In each round m queries are made. The N rounds are divided into 3 phases.

Phase 1 ($1 \leq i \leq 0.4N$): In the first round m queries are made, for which the bidders $0.1N + 1$ to N bid with a bid of a , and bidders 1 to $0.1N$ do not bid. Similarly, for $1 \leq i \leq 0.4N$, in the i th round m queries are made, for which bidders $0.1N + i$ to N bid with a bid of a , and for which bidders 1 to $0.1N + i - 1$ do not bid.

For $1 \leq i \leq 0.4N$, the algorithm will distribute the queries of the i th round equally between bidders $0.1N + i$ to N . This will give the partial allocation as shown in Figure 2.

Phase 2 ($0.4N + 1 \leq i \leq 0.5N$): In the $(0.4N + 1)$ th round m queries are made, for which bidder 1 bids a , and bidders $0.5N$ to N bid $2a$ (the rest of the bidders bid 0). Similarly, for $0.4N + 1 \leq i \leq 0.5N$, in the i th round m queries are made, for which bidder $i - 0.4N$ bids a , and bidders $0.5N$ to N bid $2a$.

For $0.4N + 1 \leq i \leq 0.5N$, the algorithm will distribute the queries of round i equally between bidders $0.5N$ to N .

At this point during the algorithm, bidders $0.5N + 1$ to N have spent all their money.

20

[form-feed in source extraction]

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.

The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent

Expanded PaperInterface specification

```
0 < EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail ^
  (∀ (owner : ℝ),
    1 / 2 ≤ owner ^ owner ≤ 1 →
      EconCSLib.Online.MSVV07Appendix.printedPhaseThreeSpentBefore
        EconCSLib.Online.MSVV07Appendix.printedPhaseThreeResidualPerTail owner =
          1) ^
    EconCSLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
      EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_three_phase_constructionSpec_proof

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 1182-1220

Verbatim source input

Figure 2: The bidders are ordered from right to left. The area inside the dark outline is the amount of money generated by the algorithm. The optimum allocation gets an amount equal to the whole rectangle.

We present an example to show a factor strictly less than $1 - 1/e$ for the algorithm which gives a query to a highest bidder, breaking ties by giving it to the bidder with most left-over money. This example has only three values for the bids - 0, a or $2a$, for some small $a > 0$. Thus, in the case of arbitrary bids, the strategy of bucketing close enough bids (say within a factor of 2) together, and running such an algorithm does not work.

There are N bidders numbered $1, \dots, N$, each with budget 1. We get the following query sequence and bidding pattern. Each bid is either 0, a or $2a$. Let $m = 1/a$. We will take $a \rightarrow 0$. The queries arrive in N rounds. In each round m queries are made. The N rounds are divided into 3 phases.

Phase 1 ($1 \leq i \leq 0.4N$): In the first round m queries are made, for which the bidders $0.1N + 1$ to N bid with a bid of a , and bidders 1 to $0.1N$ do not bid. Similarly, for $1 \leq i \leq 0.4N$, in the i th round m queries are made, for which bidders $0.1N + i$ to N bid with a bid of a , and for which bidders 1 to $0.1N + i - 1$ do not bid.

For $1 \leq i \leq 0.4N$, the algorithm will distribute the queries of the i th round equally between bidders $0.1N + i$ to N . This will give the partial allocation as shown in Figure 2.

Phase 2 ($0.4N + 1 \leq i \leq 0.5N$): In the $(0.4N + 1)$ th round m queries are made, for which bidder 1 bids a , and bidders $0.5N$ to N bid $2a$ (the rest of the bidders bid 0). Similarly, for $0.4N + 1 \leq i \leq 0.5N$, in the i th round m queries are made, for which bidder $i - 0.4N$ bids a , and bidders $0.5N$ to N bid $2a$.

For $0.4N + 1 \leq i \leq 0.5N$, the algorithm will distribute the queries of round i equally between bidders $0.5N$ to N .

At this point during the algorithm, bidders $0.5N + 1$ to N have spent all their money.

20

[form-feed in source extraction]

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.

The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent their money.

The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$ amount of money. Thus the factor is strictly less than $1 - 1/e$.

Expanded PaperInterface specification

```
EconCSLib.Online.MSVV07Appendix.printedContinuousOnlineValue = 9 / 10 - 1 / 2 * Real.log (9 / 5) ^
EconCSLib.Online.MSVV07Appendix.printedPhaseThreeServedMass = 3 / 10 - 1 / 2 * Real.log (9 / 5) ^
0 < EconCSLib.Online.MSVV07Appendix.printedPhaseThreeServedMass ^ 2 / 5 + 2 * (1 / 10) = 3 / 5 ^ 3 / 5 ≠ 31 / 50
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_algorithm_revenue_formulaSpec_proof

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 1210-1219

Verbatim source input

Phase 3 ($0.5N + 1 \leq i \leq N$): m queries enter in round i , for which only bidder i bids at a , and the other bidders do not bid.
The algorithm has to throw away these queries, since bidders $0.5N + 1$ to N have already spent their money.
The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$

Expanded PaperInterface specification

```
((∀ (r : ℝ), 0 ≤ r → r ≤ 2 / 5 → EconCSLib.Online.MSVV07Appendix.printedPhaseOneRawBid r (1 / 10 + r) = 1) ∧
(∀ (offset : ℝ),
  0 ≤ offset → offset ≤ 1 / 10 → EconCSLib.Online.MSVV07Appendix.printedPhaseTwoRawBid offset offset = 1) ∧
(∀ (owner : ℝ),
  1 / 2 ≤ owner → owner ≤ 1 → EconCSLib.Online.MSVV07Appendix.printedPhaseThreeRawBid owner owner = 1) ∧
  2 / 5 + 1 / 10 + 1 / 2 = 1) ∧
(EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun bidder => 1) ∧
EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue = 1 ∧
  ∀ (spend : ℝ → ℝ),
    EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible spend →
      ∫ (bidder : ℝ) in 0..1, spend bidder ≤ EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_optimum_revenue_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 1215-1221

Verbatim source input

The optimum allocation, on the other hand, is to allocate the queries in round i as follows:

- For $1 \leq i \leq 0.4N$, allocate all queries in round i to bidder $0.1N + i$.
- For $0.4N + 1 \leq i \leq 0.5N$, allocate all queries in round i to bidder $i - 0.4N$.
- For $0.5N + 1 \leq i \leq N$, allocate all queries in round i to bidder i .

Clearly, OPT makes N amount of money. A calculation shows that the algorithm makes $0.62N$ amount of money. Thus the factor is strictly less than $1 - 1/e$.
We can modify the above example to allow bids of θ , a and κa , for any $\kappa > 1$, such that the

Expanded PaperInterface specification

```
EconCSLib.Online.MSVV07Appendix.printedContinuousOnlineValue /  
  EconCSLib.Online.MSVV07Appendix.printedContinuousOfflineValue <  
  EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Lean proof endpoint (built separately)

```
EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_naive_strict_ratio_conclusionSpec_proof
```

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 38

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of θ , a and k_a , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$.
As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Expanded PaperInterface specification

```

∀ {κ : ℝ},
  1 < κ →
    EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue κ =
      EconCSLib.Online.MSVV07Appendix.derivedKappaRatio κ ∧
      EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue κ = 1 ∧
      (EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible fun _bidder => 1) ∧
      (∀ (spend : ℝ → ℝ),
        EconCSLib.Online.MSVV07Appendix.ContinuousUnitBudgetFeasible spend →
          ∫ (bidder : ℝ) in 0..1, spend bidder ≤
            EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue κ) ∧
      EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue κ /
        EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue κ <
          EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_continuous_source_execution

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 39

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of θ , a and k_a , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$. As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Expanded PaperInterface specification

```
(∀ (κ : ℝ),
  1 < κ →
    EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue κ /
      EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue κ =
        EconCSLib.Online.MSVV07Appendix.derivedKappaRatio κ) ∧
  Filter.Tendsto EconCSLib.Online.MSVV07Appendix.derivedKappaRatio Filter.atTop
    (nhds EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio)
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_ratio_limit_infinity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 40

Source locator: cited publication, lines 1221-1224

Verbatim source input

We can modify the above example to allow bids of θ , a and k_a , for any $\kappa > 1$, such that the algorithm performs strictly worse than $1 - 1/e$.
As $\kappa \rightarrow \infty$, the factor tends to $1 - 1/e$, and as $\kappa \rightarrow 1$, the factor tends to $1/2$. Of course, if $\kappa = 1$, then this reduces to the original model of [KP00], and the factor is $1 - 1/e$.

Expanded PaperInterface specification

```
(∀ (κ : ℝ),
  1 < κ →
    EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOnlineValue κ /
      EconCSLib.Online.MSVV07Appendix.derivedKappaContinuousOfflineValue κ =
        EconCSLib.Online.MSVV07Appendix.derivedKappaRatio κ) ∧
  Filter.Tendsto EconCSLib.Online.MSVV07Appendix.derivedKappaRatio (nhdsWithin 1 (Set.Ioi 1)) (nhds (1 / 2)) ∧
    EconCSLib.Online.MSVV07Appendix.derivedKappaRatio 1 = 1 / 2 ∧
    1 / 2 ≠ EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.appendix_kappa_ratio_limit_one

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 41

Source locator: cited publication, lines 897-914

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/[U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N , by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D .

Expanded PaperInterface specification

$\forall (N : \mathbb{N}),$
EconCSLib.Online.MSVV07PaperFacing.theorem9HardDistribution $N =$ EconCSLib.Online.uniformPermutationDistribution N

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9HardDistribution_uniform

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 42

Source locator: cited publication, lines 897-918

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/[U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N , by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D . Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:
(
1

Expanded PaperInterface specification

```
∀ (N m : ℕ) (epsilon : ℝ) (permutation : Equiv.Perm (Fin N)),
  ↑m * epsilon = 1 →
    (∀ (round bidder : Fin N),
      EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9RoundBid N epsilon permutation round bidder =
        if ↑round ≤ ↑bidder then epsilon else 0) ∧
      EconCSLib.Online.MSVV07PaperFacing.paperRevenue
        (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardInstance N m epsilon permutation)
        (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9HardOfflineAssignment N m permutation) =
        ↑N
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_hard_instance_round_bids_and_optimum

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 43

Source locator: cited publication, lines 897-908

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b.
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/(U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i, which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i, while bidders 1 through i - 1 bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i, achieving a revenue of N. One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input. Now consider all the inputs which can be derived from the above input by permutation of the

Expanded PaperInterface specification

```

∀ (delta : ℝ),
  0 < delta →
    ∃ N0,
      ∀ (N : ℕ),
        N0 ≤ N →
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9BaseEqualSpreadBalanceRevenue N / ↑N ≤
            EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_base_instance_balance_revenue_claim

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 44

Source locator: cited publication, lines 918-931

Verbatim source input

1
if $j \geq i$,
E $\pi [q_{ij}] \leq N^{-i+1}$
0
if $j < i$.
To see this, note that there are $N - i + 1$ bidders who are bidding for queries Q_i . The deterministic algorithm allocates some fraction of these queries to some bidders who bid for them, and leaves the rest of the queries unallocated. If $j \geq i$ then bidder j is a random bidder among the bidders bidding for these queries and hence is allocated an average amount 1 of the queries which were allocated from Q_i (where the average is taken over random of N^{-i+1} permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation.

Expanded PaperInterface specification

$\forall (N : \mathbb{N})$ (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm N) (round bidder : Fin N),
(EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder $\leq$
if $\uparrow \text{round} \leq \uparrow \text{bidder}$ then $1 / \uparrow (N - \uparrow \text{round})$ else 0) $\wedge$
($\neg \uparrow \text{round} \leq \uparrow \text{bidder} \rightarrow$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder = 0)

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_expected_round_allocation_bound

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 45

Source locator: cited publication, lines 929-938

Verbatim source input

of $N - i + 1$ permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation. Thus we get that the amount of money spent by a bidder j at the end of the algorithm P_j^{expected} is at most $\min\{1, \sum_{i=1}^N \frac{1}{N - i + 1}\}$. By summing this over $j = 1, \dots, N$, we get that the expected revenue of the deterministic algorithm over the distributional input D is at most $N(1 - 1/e)$. This finishes the proof of the theorem.

Expanded PaperInterface specification

```

∀ (N : ℕ) (algorithm : EconCSLib.Online.BMatchingIntegralPrefixAlgorithm N) (bidder : Fin N),
  EconCSLib.Online.theorem9BidderSpendUpperBound N bidder =
    min 1 (∑ round, if ↑round ≤ ↑bidder then 1 / ↑(N - ↑round) else 0) ∧
    min 1 (
      ∑ round,
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9ExpectedRoundAllocation N algorithm round bidder) ≤
      EconCSLib.Online.theorem9BidderSpendUpperBound N bidder ∧
    EconCSLib.Online.theorem9NormalizedRevenueUpperBound N =
      (∑ b, EconCSLib.Online.theorem9BidderSpendUpperBound N b) / ↑N ∧
    (EconCSLib.pmfExp (EconCSLib.Online.uniformPermutationDistribution N) fun permutation =>
      (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.theorem9IntegralRoundCertificate N).normalizedRevenue
        algorithm permutation) ≤
      EconCSLib.Online.theorem9NormalizedRevenueUpperBound N ∧
  ∀ (delta : ℝ),
    0 < delta →
      ∃ N0,
        ∀ (marketSize : ℕ),
          N0 ≤ marketSize →
            EconCSLib.Online.theorem9NormalizedRevenueUpperBound marketSize ≤
              EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_harmonic_spend_and_revenue_bound

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 46

Source locator: cited publication, lines 915-938

Verbatim source input

Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:

(
1
if $j \geq i$,
En $[q_{ij}] \leq N^{-i+1}$
0
if $j < i$.
To see this, note that there are $N - i + 1$ bidders who are bidding for queries Q_i . The deterministic algorithm allocates some fraction of these queries to some bidders who bid for them, and leaves the rest of the queries unallocated. If $j \geq i$ then bidder j is a random bidder among the bidders bidding for these queries and hence is allocated an average amount
1
of the queries which were allocated from Q_i (where the average is taken over random
of N^{-i+1}
permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation.
Thus we get that the
amount of money spent by a bidder j at the end of the algorithm
 P_j^{expected}
1
is at most $\min\{1, N^{-i+1}\}$. By summing this over $j = 1, \dots, N$, we get that the expected revenue of the deterministic algorithm over the distributional input D is at most $N(1 - 1/e)$.
This finishes the proof of the theorem.

Expanded PaperInterface specification

```
∀ (N : ℕ) (algorithm : { choice // choice.Feasible }) (permutation : Equiv.Perm (Fin N)),
  EconCSLib.Online.MSVV07PaperFacing.theorem9CappedNormalizedRevenue N algorithm permutation =
    (∑ bidder,
      min 1
        (∑ round,
          EconCSLib.Online.BMatchingIntegralPrefixAlgorithm.prefixAllocation algorithm
            (EconCSLib.Online.theorem90bservedPrefix N permutation round) round (permutation bidder))) /
    ↑N
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9CappedNormalizedRevenue_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 47

Source locator: cited publication, lines 897-938

Verbatim source input

Theorem 9 No randomized online algorithm can have a competitive ratio better than $1 - 1/e$ for the b-matching problem, for large b .
Proof : By Yao's Lemma [Yao77], it suffices to present a distribution over inputs such that any deterministic algorithm obtains at most $1 - 1/e$ of the optimal allocation on the average. Consider first the worst case input for the algorithm BALANCE with N bidders, each with a budget of 1. In this instance, the queries enter in N rounds, with $1/[U+000F$ control character] number of queries in each round. We denote by Q_i the queries of round i , which are identical to each other. For every $i = 1, \dots, N$, bidders i through N bid $[U+000F$ control character] for each of the queries of round i , while bidders 1 through $i - 1$ bid 0 for these queries. The optimal assignment is clearly the one in which all the queries of round i are allocated to bidder i , achieving a revenue of N . One can show that BALANCE will achieve only $N(1 - 1/e)$ revenue on this input.
Now consider all the inputs which can be derived from the above input by permutation of the numbers of the bidders and take the uniform distribution D over all these inputs. Formally, D can be described as follows: Pick a random permutation π of the bidders. The queries enter in rounds in the order $Q_1, Q_2, \dots, Q_N$. Bidders $\pi(i), \pi(i + 1), \dots, \pi(N)$ bid $[U+000F$ control character] for the queries Q_i and the other bidders bid 0 for these queries. The optimal allocation for any permutation π remains N , by allocating the queries Q_i to bidder $\pi(i)$. We wish to bound the expected revenue of any deterministic algorithm over inputs from the distribution D .
Fix any deterministic algorithm. Let q_{ij} be the fraction of queries from Q_i that bidder j is allocated. We have:
(
1
if $j \geq i$,
 $E\pi[q_{ij}] \leq N - i + 1$
0
if $j < i$.
To see this, note that there are $N - i + 1$ bidders who are bidding for queries Q_i . The deterministic algorithm allocates some fraction of these queries to some bidders who bid for them, and leaves the rest of the queries unallocated. If $j \geq i$ then bidder j is a random bidder among the bidders bidding for these queries and hence is allocated an average amount 1 of the queries which were allocated from Q_i (where the average is taken over random of $N - i + 1$ permutations of the bidders). On the other hand, if $j < i$, then bidder j bids 0 for queries in Q_i and is not allocated any of these queries in any permutation.
Thus we get that the amount of money spent by a bidder j at the end of the algorithm
 P_j expected
1
is at most $\min\{1, i = 1 \dots N - i + 1\}$. By summing this over $j = 1, \dots, N$, we get that the expected revenue of the deterministic algorithm over the distributional input D is at most $N(1 - 1/e)$. This finishes the proof of the theorem.

Expanded PaperInterface specification

```
∀ (delta : ℝ),
  0 < delta →
    ∃ N0,
      ∀ (N : ℕ),
        N0 ≤ N →
          ∀ (randomizedAlgorithm : EconCSLib.Online.MSVV07PaperFacing.theorem9RandomizedOnlineAlgorithm N),
            ¬∀ (permutation : Equiv.Perm (Fin N)),
              EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio + delta <
                EconCSLib.pmfExp randomizedAlgorithm fun algorithm =>
                  EconCSLib.Online.MSVV07PaperFacing.theorem9CappedNormalizedRevenue N algorithm permutation
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem9_no_randomized_online_algorithm_beats_msvv_ratio

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 48

Source locator: cited publication, lines 247-264

Verbatim source input

In this section we analyze the performance of our algorithm in the special case when all bids are equal. This is exactly the algorithm BALANCE of [KP00]. We give a simpler analysis of this algorithm using the notion of a factor-revealing LP. This technique was implicit in [MRJW77, GK98, MMSV01] and was formalized and made explicit in [JMS02, JMM+ 03]. We will see how to extend the analysis to the general case in Section 5. For another simple proof for BALANCE see [AL06].

We will assume for simplicity that in the optimum solution, each of the N players spends his entire budget, and thus the total revenue is N (the proof is similar even without this assumption, and we provide it in Section 6). Recall that BALANCE awards each query to the interested bidder who has the maximum unspent budget. We wish to lower bound the total revenue achieved by BALANCE. Let us define the type of a bidder according to the fraction of budget spent by that bidder at the end of the algorithm BALANCE: say that the bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j - 1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. Clearly bidders of type j for small values of j contribute little to the total revenue. The factor revealing LP for the performance of the algorithm BALANCE will proceed by bounding the number of such bidders of type j .

Lemma 1 If OPT assigns query q to a bidder B of type $j \leq k - 1$, then BALANCE pays for

Expanded PaperInterface specification

```
∀ (k : ℕ) (spentFraction : ℝ) (i : Fin k),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k spentFraction i ↔
    spentFraction = 0 ∧ i = 0 ∨ i = 1 / k < spentFraction ∧ spentFraction ≤ i(i + 1) / k
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_bidder_type_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 49

Source locator: cited publication, lines 273-279

Verbatim source input

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} [\text{inst} : \text{Fintype Advertiser}] (k : \mathbb{N}) (\text{finalSpentFraction} : \text{Advertiser} \rightarrow \mathbb{R}) (i : \text{Fin } k),$
 $\text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4TypeCount } k \text{ finalSpentFraction } i =$
 $\sum a, \text{ if EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType } k (\text{finalSpentFraction } a) \text{ then } 1 \text{ else } 0$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_type_count_x_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 50

Source locator: cited publication, lines 273-300

Verbatim source input

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

1

3/k
SLAB 3
2/k
TYPE 2

1/k
0

xk
x3
x2
x1

Figure 1: The bidders are ordered from right to left in order of increasing type. We have labeled here the bidders of type 2 and the money in slab 3.
Lemma 2

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (N : \mathbb{R}) (x : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section4IdealizedSlabSpend $N \times i =$
 $N / \uparrow k - (\sum j \in \text{Finset.Iio } i, x \ j) / \uparrow k$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_slab_spend_beta_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 51

Source locator: cited publication, lines 270-279

Verbatim source input

for allocating q , and therefore B must allocate q to some bidder who has spent at most j/k fraction of his budget.
For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k - 1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

Expanded PaperInterface specification

```

∀ {Advertiser : Type u_5} [inst : Fintype Advertiser] {k : ℕ},
  0 < k →
    ∀ (error : Advertiser → ℝ),
      (∀ (a : Advertiser), 0 ≤ error a) →
        (∀ (a : Advertiser), error a ≤ 1 / ↑k) → ∑ a, error a ≤ ↑(Fintype.card Advertiser) / ↑k
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_discretization_error_bound

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 52

Source locator: cited publication, lines 344-456

Verbatim source input

The revenue of the algorithm is
 $BAL \geq$

$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k x_i +$$
$$\sum_{i=1}^k$$
$$= N -$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^k$$

To find a P lower bound on the performance of BALANCE we want to find the minimum value

$$\sum_{i=1}^{k-1} \sum_{j=1}^k$$
$$\sum_{i=1}^k$$

that $\sum_{i=1}^k x_i - k$ can take over the feasible $\{x_i\}$ s. This gives the following LP, which

[form-feed in source extraction]

we call L. In both the constraints below, i ranges from 1 to $k - 1$.

maximize

$$\Phi =$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$
$$\sum_{i=1}^{k-1}$$
$$\sum_{i=1}^k$$

```

k-1
X
k-i
j-i
) yj ≥
(1 +
k
k
j=i

∀ i : yi ≥ 0

Define A, b, c so the primal LP, L, can be written as
max c · x

s.t. Ax ≤ b x ≥ 0.

and the dual LP, D, can be written as
min b · y

s.t. AT y ≥ c

y ≥ 0.

Lemma 3 As k → ∞, the value ϕ of the linear programs L and D goes to Ne

```

Expanded PaperInterface specification

```

∀ {m : ℕ} (N : ℝ) (x y : Fin m → ℝ),
  (EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).primalObjective x =
    EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective x ∧
  ((EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).PrimalFeasible x ↔
    EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible N x) ∧
  (EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).dualObjective y =
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualObjective N y ∧
  ((EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m N).DualFeasible y ↔
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualFeasible y)

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_factor_revealing_primal_dual_lp_formulas

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 53

Source locator: cited publication, lines 456-490

Verbatim source input

Lemma 3 As $k \rightarrow \infty$, the value Φ of the linear programs L and D goes to $N\epsilon$
Proof : On setting all the primal constraints to equality and solving the resulting system, we get a feasible solution $x_i \geq 0$. Similarly, we can set all the dual constraints to equality and solve the resulting system to get a feasible dual solution.
These two feasible solutions are:
 $x_i = Nk(1 - k_1)^{i-1}$

for $i = 1, \dots, k - 1$

 $y_i = k_1(1 - k_1)^{k-i-1}$

for $i = 1, \dots, k - 1$

Clearly they satisfy all complementary slackness conditions, hence they are also optimal solutions of the primal and dual programs.
8

[form-feed in source extraction]
This gives an optimal objective function value of
 $\Phi = c \cdot x^* = b \cdot y^*$
=

k-1
X

N
1 i-1
(k-i
k) k (1 - k)

i=1

= N (1 - k₁)^k
As we make the discretization finer (i.e. as $k \rightarrow \infty$) Φ tends to $N\epsilon$.
[U+0003 control character]
Recall that the size of the matching is at least $N - \Phi - Nk$, hence it tends to $N(1 - \epsilon)$. Since OPT is N , the competitive ratio is at least $1 - \epsilon$.

Expanded PaperInterface specification

```
∀ {m N : ℕ},
  EconCSLib.Optimization.IsMaximizerOn (EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalFeasible ↑N)
    EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective
    (EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate ↑N) ∧
  EconCSLib.Optimization.IsMinimizerOn (EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m ↑N).DualFeasible
    (EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLP m ↑N).dualObjective
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate ∧
  EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalObjective
    (EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrimalCandidate ↑N) =
    EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue m ↑N ∧
  EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualObjective (↑N)
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteDualCandidate =
    EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue m ↑N
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemma3_displayed_primal_dual_witnesses_optimal

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 54

Source locator: cited publication, lines 264-272

Verbatim source input

Lemma 1 If OPT assigns query q to a bidder B of type $j \leq k - 1$, then BALANCE pays for q from some slab i such that $i \leq j$.
The lemma follows immediately from the criterion used by BALANCE for assigning queries to bidders: B has type $j \leq k - 1$ and therefore spends at most $j/k < 1$ fraction of his budget at the end of BALANCE. It follows that when query q arrives, B is available to BALANCE

[form-feed in source extraction]
for allocating q , and therefore B must allocate q to some bidder who has spent at most j/k fraction of his budget.

Expanded PaperInterface specification

```

∀ {Slab : Type u_5} [inst : LinearOrder Slab] (psi : Slab → ℝ) {optType optCurrentSlab chosenSlab : Slab}
  {bid chosenBid : ℝ},
  optCurrentSlab ≤ optType →
    bid * psi optCurrentSlab ≤ chosenBid * psi chosenSlab →
      chosenBid = bid → 0 < bid → StrictAnti psi → chosenSlab ≤ optType

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemmal_balance_pays_no_later_slab

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 55

Source locator: cited publication, lines 273-330

Verbatim source input

For simplicity we will assume that bidders of type i spend exactly i/k fraction of their budget, and that queries do not straddle slabs. The latter is justified by the fact that bids are small compared to budgets (e.g. taking bids to be smaller than $k/2$ of the budget). The total error resulting from this simplification is at most N/k and is negligible, once we take k to be large enough. Now, for $i = 1, 2, \dots, k-1$, let x_i be the number of bidders of type (i) . Let β_i denote the total money spent by the bidders from slab i in the run of BALANCE. It is easy to see (Figure 1) that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (x_1 + \dots + x_{i-1})/k$.

1
3/k
SLAB 3
2/k
TYPE 2

1/k
0

x_k

x_3

x_2

x_1

Figure 1: The bidders are ordered from right to left in order of increasing type. We have labeled here the bidders of type 2 and the money in slab 3.

Lemma 2
$$\sum_{i=1}^k (1 + \frac{i}{k}) x_i \leq N$$

$\forall i, 1 \leq i \leq k-1 :$

$j=1$

Proof :

By Lemma 1,
$$\sum_{i=1}^k x_i \leq$$

$x_j \leq$

$$\sum_{j=1}^i x_j$$

$j=1$

i

Expanded PaperInterface specification

$$\forall \{m : \mathbb{N}\} (N : \mathbb{R}) (x \text{ beta} : \text{Fin } m \rightarrow \mathbb{R}) (i : \text{Fin } m),$$
$$\sum_{j \in \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRoutePrefix } i, x_j \leq} \sum_{j \in \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRoutePrefix } i, \text{beta } j \rightarrow} \sum_{j \in \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRoutePrefix } i, \text{beta } j =} \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRouteRhs } N \text{ } i - \sum_{j \in \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRoutePrefix } i, \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRouteDeltaCoeff } i \text{ } j * x_j \rightarrow} \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRouteLPRow } x \text{ } i \leq \text{EconCSLib.Offline.MSVV07SourceLemmas.paperRouteRhs } N \text{ } i$$

Lean proof endpoint (built separately)

EconCSLib.Offline.MSVV07PaperFacing.PaperInterface.section4_lemma2_factor_revealing_lp_constraint

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 56

Source locator: cited publication, lines 456-490

Verbatim source input

Lemma 3 As $k \rightarrow \infty$, the value Φ of the linear programs L and D goes to $N\epsilon$
Proof : On setting all the primal constraints to equality and solving the resulting system, we get a feasible solution $x_i \geq 0$. Similarly, we can set all the dual constraints to equality and solve the resulting system to get a feasible dual solution.
These two feasible solutions are:
 $x_i = Nk (1 - k_1)^{i-1}$

for $i = 1, \dots, k - 1$

 $y_i = k_1 (1 - k_1)^{k-i-1}$

for $i = 1, \dots, k - 1$

Clearly they satisfy all complementary slackness conditions, hence they are also optimal solutions of the primal and dual programs.
8

[form-feed in source extraction]
This gives an optimal objective function value of
 $\Phi = c \cdot x^* = b \cdot y^*$
=

 $\sum_{i=1}^{k-1} Nk (1 - k_1)^{i-1}$

 $\sum_{i=1}^{k-1} k_1 (1 - k_1)^{k-i-1}$

 $= N (1 - k_1)^k$
As we make the discretization finer (i.e. as $k \rightarrow \infty$) Φ tends to $N\epsilon$.
[U+0003 control character]
Recall that the size of the matching is at least $N - \Phi - Nk$, hence it tends to $N (1 - \epsilon)$. Since OPT is N , the competitive ratio is at least $1 - \epsilon$.

Expanded PaperInterface specification

$\forall (N : \mathbb{R}), \text{EconCSLib.Sequence.SeqTendsTo } (\text{fun } k \Rightarrow N * (1 - 1 / \uparrow k) ^ k) (N / \text{Real.exp } 1)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_lemma3_factor_revealing_lp_optimality

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 57

Source locator: cited publication, lines 488-495

Verbatim source input

[U+0003 control character]
Recall that the size of the matching is at least $N - \Phi - Nk$, hence it tends to $N(1 - 1e)$. Since OPT is N , the competitive ratio is at least $1 - 1e$.
On the other hand one can find an instance of the problem (e.g., the one provided in [KP00]) such that at the end of the algorithm all the inequalities of the primal are tight, hence the competitive ratio of `BALANCE` is exactly $1 - 1e$.

5

Expanded PaperInterface specification

```
∀ (m : ℕ) (N : ℝ),
  0 ≤ N →
  have E := EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFluidExecution m N;
  (∀ (i : Fin m), 0 ≤ E.typeMass i) ∧
  (∀ (i : Fin m), E.activeMassBeforeSlab i - E.activeMassBeforeSlab (i + 1) = E.typeMass i) ∧
  (∀ stage ≤ m,
    E.spentBeforeSlab stage = i stage / i(m + 1) ∧
    E.allocationPerActiveBidder stage = 1 / i(m + 1) ∧
    E.queryMassAtSlab stage = E.activeMassBeforeSlab stage / i(m + 1) ∧
    (stage < m →
      E.spentBeforeSlab (stage + 1) = E.spentBeforeSlab stage + E.allocationPerActiveBidder stage) ∧
      E.spentBeforeSlab stage + E.allocationPerActiveBidder stage ≤ 1 ∧
      E.activeMassBeforeSlab stage * E.allocationPerActiveBidder stage = E.queryMassAtSlab stage) ∧
    ∑ i, E.typeMass i + E.fullMass = N ∧
    (∀ (i : Fin m),
      EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow E.typeMass i =
      EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i) ∧
    E.revenue =
      ∑ i,
        E.typeMass i *
        EconCSLib.Online.MSVV07PaperFacing.SourceRunner.factorLPTightFinalSpendFraction m i +
        E.fullMass ∧
    E.revenue =
      ∑ stage ∈ Finset.range (m + 1), E.activeMassBeforeSlab stage * E.allocationPerActiveBidder stage ∧
    E.revenue = N - EconCSLib.Online.MSVV07SourceLemmas.factorRevealingLPValue m N
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section4_balance_exact_tightness_instance

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 58

Source locator: cited publication, lines 509-549

Verbatim source input

Here is how we proceed instead: For every monotonically decreasing tradeoff function ψ and every instance π of the adwords problem and write a new LP $L(\pi, \psi)$ for our algorithm using tradeoff function ψ run on the instance π . Of course, once we specify the algorithm as well as the input instance, the actual allocations of queries to bidders is completely determined. In particular, the number α_i of bidders of type i is fixed. $L(\pi, \psi)$ is the seemingly trivial LP obtained by taking the left hand side of each inequality in the factor revealing LP L and substituting $x_i = \alpha_i$ to obtain the right hand side. Formally:

Recall the LP L from the previous section:

$\max c \cdot x$

s.t. $Ax \leq b$ $x \geq 0$

Let a be a $k - 1$ dimensional vector whose i th component is α_i . Let $Aa = l$. We denote the following LP by $L(\pi, \psi)$:

$\max c \cdot x$

s.t. $Ax \leq l$

$x \geq 0$
9

[form-feed in source extraction]

The dual LP is denoted by $D(\pi, \psi)$ and is:

$\min l \cdot y$

s.t. $ATy \geq c$

$y \geq 0$

Clearly, any one LP $L(\pi, \psi)$ offers no insight into the performance of our algorithm; after all the right hand sides of the inequalities are expressed in terms of the unknown number of bidders of type i . Nevertheless, the entire family $L(\pi, \psi)$ does contain useful information which is revealed by considering the duals of these LP's.

Since $L(\pi, \psi)$ differs from L only in the right hand side, the dual $D(\pi, \psi)$ differs from D only in the dual objective function; the constraints remain unchanged. Hence solution y^* of D is feasible for $D(\pi, \psi)$ as well. Recall that this solution was obtained by setting all nontrivial inequalities of D to equality.

Now by construction, if we set all the nontrivial inequalities of LP $L(\pi, \psi)$ to equality we get a feasible solution, namely a . Clearly, a and y^* satisfy all complementary slackness conditions.

Therefore they are both optimal. Hence we get:

Lemma 4 For any instance π and monotonically decreasing tradeoff function ψ , y^* is an optimal solution to $D(\pi, \psi)$.

Expanded PaperInterface specification

```
∀ {m : ℕ} (c l x y : Fin m → ℝ),
  (EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP c l).primalObjective x = ∑ i, c i * x i ∧
  ((EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP c l).PrimalFeasible x ↔
    (∀ (i : Fin m), 0 ≤ x i) ∧
    ∀ (r : Fin m), ∑ i, EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff r i * x i ≤ l r) ∧
  (EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP c l).dualObjective y = ∑ r, y r * l r ∧
  ((EconCSLib.Online.MSVV07SourceLemmas.tradeoffRevealingLP c l).DualFeasible y ↔
    (∀ (r : Fin m), 0 ≤ y r) ∧
    ∀ (i : Fin m), c i ≤ ∑ r, EconCSLib.Online.MSVV07SourceLemmas.paperRouteMatrixCoeff r i * y r)
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_tradeoff_revealing_primal_dual_lp_formulas

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 59

Source locator: cited publication, lines 550-563

Verbatim source input

The structure of the algorithm does constrain how the LP L differs from $L(\pi, \psi)$. This is what we will explore now.
As in the analysis of BALANCE, we divide the budget of each bidder into k equal slabs, numbered 1 to k . Money in slab i is spent before moving to slab $i + 1$. We say that a bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j - 1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. As before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.
Lemma 5
Proof :

Expanded PaperInterface specification

$\forall (k : \mathbb{N}) (spentFraction : \mathbb{R}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.IsFinalType k $spentFraction\ i \leftrightarrow$
 $spentFraction = 0 \wedge \forall i = 0 \vee i+1 / i+k < spentFraction \wedge spentFraction \leq i(i+1) / i+k$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_bidder_type_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 60

Source locator: cited publication, lines 550-569

Verbatim source input

The structure of the algorithm does constrain how the LP L differs from $L(\pi, \psi)$. This is what we will explore now.

As in the analysis of BALANCE, we divide the budget of each bidder into k equal slabs, numbered 1 to k . Money in slab i is spent before moving to slab $i + 1$. We say that a bidder is of type j if the fraction of his budget spent at the end of the algorithm lies in the range $((j - 1)/k, j/k]$. By convention a bidder who spends none of his budget is assigned type 1. As before, we make the simplifying assumption (at the cost of a negligible error term) that bidders of type j spend exactly j/k fraction of their budget. Let α_j denote the number of bidders of type j . Let β_i denote the total money spent by the bidders from slab i in the run of the algorithm. It is easy to see that $\beta_1 = N/k$, and for $2 \leq i \leq k$, $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$. Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$. The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.

Lemma 5

Proof :

$l = b + \Delta(\pi, \psi)$.

Consider the i th components of the three vectors. We need to prove:

$i-2$

$\alpha_1 (1 + i-1$

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (N : \mathbb{R}) (\alpha : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta $N \alpha i =$
 $N / \uparrow k - (\sum j \in \text{Finset.Iio } i, \alpha j) / \uparrow k$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_beta_slab_spend_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 61

Source locator: cited publication, lines 566-577

Verbatim source input

Consider the i th components of the three vectors. We need to prove:

$$\begin{aligned} & \alpha_1 \frac{1 + i - 1}{k} + \alpha_2 \frac{1 + k}{k} + \dots + \alpha_i \\ &= \frac{iN}{k} + (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i). \end{aligned}$$

This equation follows using the fact that $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$.

[U+0003 control character]

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (\alpha \ \beta : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5Delta $\alpha \ \beta \ i = \sum j \in \text{Finset.Iic } i, (\alpha \ j - \beta \ j)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_delta_prefix_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 62

Source locator: cited publication, lines 579-586

Verbatim source input

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]

Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (d : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue I d =
    match d.winner with
    | none => 0
    | some a => I.bid a d.query
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_alg_query_revenue_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 63

Source locator: cited publication, lines 579-586

Verbatim source input

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]
Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .
Lemma 6 For each query q such that $1 \leq type(q) \leq k - 1$,

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (history : List Query) (opt : Fin history.length → Option Advertiser) (t : Fin history.length),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section50ptQueryRevenue I history opt t =
    match opt t with
    | none => 0
    | some a => I.bid a (history.get t)
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_opt_query_revenue_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 64

Source locator: cited publication, lines 579-586

Verbatim source input

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]
Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .
Lemma 6 For each query q such that $1 \leq type(q) \leq k - 1$,

Expanded PaperInterface specification

$\forall \{Advertiser : Type\ u_1\} \{k : \mathbb{N}\} (optOwner : Option\ Advertiser) (finalType : Advertiser \rightarrow Fin\ k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QueryType optOwner finalType = Option.map finalType optOwner

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_query_type_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 65

Source locator: cited publication, lines 579-586

Verbatim source input

We are interested in comparing the performance of our algorithm (abbreviated as ALG) with the optimal algorithm OPT. The following definitions focus on some relevant parameters comparing how ALG and OPT treat a query q :

[form-feed in source extraction]
Definition: Let $ALG(q)$ ($OPT(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .
Lemma 6 For each query q such that $1 \leq type(q) \leq k - 1$,

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (d : EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk I d =
    Option.map
      (fun a => EconCSLib.Online.MSVV07PaperFacing.SourceRunner.activeBudgetSlab k hk (d.spentBefore a) (I.budget a))
      d.winner
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_query_slab_definition

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 66

Source locator: cited publication, lines 597-617

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq \sum_{i=1}^k$$

Proof :

$\sum_{i=1}^k$

We start by observing that for $1 \leq i \leq k - 1$:

$$\sum_{i=1}^k$$

$OPT(q) = \sum_{i=1}^k \alpha_i$

$$\sum_{i=1}^k$$

$ALG(q) = \sum_{i=1}^k \beta_i$

Expanded PaperInterface specification

$\forall \{ \text{Advertiser} : \text{Type } u_5 \} \{ \text{Query} : \text{Type } u_6 \} [\text{inst} : \text{Fintype Advertiser}] [\text{inst}_1 : \text{DecidableEq Advertiser}] \{k : \mathbb{N}\}$
 $(I : \text{EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query}) (\text{history} : \text{List Query})$
 $(\text{opt} : \text{Fin history.length} \rightarrow \text{Option Advertiser}) (\text{finalType} : \text{Advertiser} \rightarrow \text{Fin } k),$
 $(\forall (a : \text{Advertiser}), \text{EconCSLib.Online.MSVV07PaperFacing.SourceRunner.occurrenceSpend } I \text{ history opt } a = 1) \rightarrow$
 $\sum_{i=1}^k (\text{if } \text{finalType } a = i \text{ then } 1 \text{ else } 0)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_type_fiber_alpha_accounting_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 67

Source locator: cited publication, lines 597-617

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq$$
$$\sum_{i=1}^k$$

Proof :

$$\sum_{i=1}^k$$

We start by observing that for $1 \leq i \leq k - 1$:

$$\sum_{i=1}^k$$
$$\text{OPT}(q) = \sum_{i=1}^k \alpha_i$$
$$\sum_{i=1}^k$$
$$\text{ALG}(q) = \sum_{i=1}^k \beta_i$$

Expanded PaperInterface specification

```
∀ {Advertiser : Type u_1} {Query : Type u_2} (k : ℕ) (hk : 0 < k)
  (I : EconCSLib.Online.MSVV07PaperFacing.PaperInstance Advertiser Query)
  (decisions : List (EconCSLib.Online.MSVV07PaperFacing.SourceRunner.OccurrenceDecision Advertiser Query)) (i : Fin k),
  EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgSlabFiberRevenue k hk I decisions i =
    (List.map
      (fun d =>
        if EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5QuerySlab k hk I d = some i then
          EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5AlgQueryRevenue I d
        else 0)
      decisions).sum
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_slab_fiber_beta_accounting_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 68

Source locator: cited publication, lines 713-731

Verbatim source input

The lemma follows from these three inequalities.
[U+0003 control character]
The final step consists of choosing the correct tradeoff function ψ as a function of the dual optimal solution y^* itself, so that for every instance π , the value of the optimal solution to $L(\pi, \psi)$ is at most that of L .
Theorem 8 For function ψ_k defined as
 $\psi_k(i) :=$
$$\begin{aligned} & k-1 \\ & X \\ & j=i \\ & 1 \\ & y_j^* = 1 - (1 -)^{k-i+1} \\ & k \end{aligned}$$

the competitive ratio of the algorithm is $(1 - 1/e)$, as k tends to infinity.
Proof :
By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $1 \cdot y^*$. By

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (i : \text{Fin } m), \text{EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiCandidate } i = 1 - (1 - 1 / (m + 1)) ^ (m - i)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_dual_induced_tradeoff_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 69

Source locator: cited publication, lines 731-773

Verbatim source input

By Lemma 4, the optimal solution to $L(\pi, \psi)$ and $D(\pi, \psi)$ has value $l \cdot y^*$. By Lemma 5 this equals $(b + \Delta) \cdot y^* \leq N/e + \Delta \cdot y^*$ (since $b \cdot y^* \leq N/e$, from Section 4). Now,

$$\begin{aligned} & \Delta \cdot y = \\ & \sum_{i=1}^{k-1} y_i ((\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)) \\ & \sum_{i=1}^{k-1} (\alpha_i - \beta_i) (y_i + \dots + y_{k-1}) \\ & \sum_{i=1}^{k-1} (\alpha_i - \beta_i) \psi(i) \\ & = \sum_{i=1}^{k-1} \psi(i) \\ & \leq \sum_{i=1}^k \psi(i) \end{aligned}$$

where the last equality follows from our choice of the function ψ , and the inequality follows from Lemma 7. As k tends to infinity, we get that the competitive ratio of our algorithm is $(1 - 1/e)$.

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} (\alpha \beta \gamma : \text{Fin } m \rightarrow \mathbb{R}),$
 $\sum_i \gamma_i \cdot \text{EconCSLib.Online.MSVV07SourceLemmas.paperRouteDelta } \alpha \beta i =$
 $\sum_i \gamma_i \cdot \text{EconCSLib.Online.MSVV07SourceLemmas.paperRoutePsiFromDual } \gamma_i \cdot (\alpha i - \beta i)$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_delta_dual_suffix_identity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 70

Source locator: cited publication, lines 777-785

Verbatim source input

The above analysis helped us derive the correct tradeoff function ψ together with the competitive ratio. However, the proof of $\hookrightarrow$ the competitive ratio of the algorithm is simpler, once we are given the correct ψ . We give a quick sketch the main steps of such a proof below:
From the definitions of the α and β variables, we have the following relations:
$$\sum_{j=1}^N \alpha_j = \sum_{i=1}^k \beta_i$$

Lemma 7 gives us

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (N : \mathbb{R}) (\alpha : \text{Fin } k \rightarrow \mathbb{R}) (i : \text{Fin } k),$
EconCSLib.Online.MSVV07PaperFacing.SourceRunner.section5IdealizedBeta N alpha i =
 $N / \uparrow k - (\sum j \in \text{Finset.Iio } i, \alpha j) / \uparrow k$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_simple_proof_beta_recurrence_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 71

Source locator: cited publication, lines 785-799

Verbatim source input

Lemma 7 gives us
X
$$\sum_i \psi(i) \alpha_i \leq$$

i
where the choice of ψ is:
X
$$\sum_i \psi(i) \beta_i$$

i

Expanded PaperInterface specification

$\forall \{k : \mathbb{N}\} (\text{psi alpha beta} : \text{Fin } k \rightarrow \mathbb{R}),$
EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8WeightedAlphaBeta psi alpha beta $\leftrightarrow$
 $\sum_i \text{psi } i * \text{alpha } i \leq \sum_i \text{psi } i * \text{beta } i$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_simple_proof_weighted_alpha_beta_inequality

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 72

Source locator: cited publication, lines 777-819

Verbatim source input

The above analysis helped us derive the correct tradeoff function ψ together with the competitive ratio. However, the proof of $\hookrightarrow$ the competitive ratio of the algorithm is simpler, once we are given the correct ψ . We give a quick sketch the main steps of such a proof below:
From the definitions of the α and β variables, we have the following relations:
$$\sum_{j=1}^N \alpha_j = \sum_{i=1}^k \beta_i$$

Lemma 7 gives us
$$\psi(i) \alpha_i \leq \sum_{j=1}^i \alpha_j$$

where the choice of ψ is:
$$\psi(i) = 1 - (1 - \frac{1}{e})^{k-i+1}$$

Combining these relations we get:
$$\sum_{i=1}^k \alpha_i \leq \sum_{i=1}^k \sum_{j=1}^i \alpha_j$$

But the left side of the inequality above is precisely the amount of money left unspent at the end of the algorithm. This establishes that the competitive ratio is $1 - 1/e$.

Expanded PaperInterface specification

$\forall (N \text{ revenue error} : \mathbb{R}),$
 $\text{EconCSLib.Online.MSVV07PaperFacing.paperMsvvRatio} * N \leq \text{revenue} + \text{error} \rightarrow N - \text{revenue} \leq N / \text{Real.exp } 1 + \text{error}$

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.theorem8_unspent_budget_bound_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 73

Source locator: cited publication, lines 509-549

Verbatim source input

Here is how we proceed instead: For every monotonically decreasing tradeoff function ψ and every instance π of the adwords problem and write a new LP $L(\pi, \psi)$ for our algorithm using tradeoff function ψ run on the instance π . Of course, once we specify the algorithm as well as the input instance, the actual allocations of queries to bidders is completely determined. In particular, the number α_i of bidders of type i is fixed. $L(\pi, \psi)$ is the seemingly trivial LP obtained by taking the left hand side of each inequality in the factor revealing LP L and substituting $x_i = \alpha_i$ to obtain the right hand side. Formally:

Recall the LP L from the previous section:

$\max c \cdot x$

s.t. $Ax \leq b$ $x \geq 0$

Let a be a $k - 1$ dimensional vector whose i th component is α_i . Let $Aa = l$. We denote the following LP by $L(\pi, \psi)$:

$\max c \cdot x$

s.t. $Ax \leq l$

$x \geq 0$

[form-feed in source extraction]

The dual LP is denoted by $D(\pi, \psi)$ and is:

$\min l \cdot y$

s.t. $ATy \geq c$

$y \geq 0$

Clearly, any one LP $L(\pi, \psi)$ offers no insight into the performance of our algorithm; after all the right hand sides of the inequalities are expressed in terms of the unknown number of bidders of type i . Nevertheless, the entire family $L(\pi, \psi)$ does contain useful information which is revealed by considering the duals of these LP's.

Since $L(\pi, \psi)$ differs from L only in the right hand side, the dual $D(\pi, \psi)$ differs from D only in the dual objective function; the constraints remain unchanged. Hence solution y^* of D is feasible for $D(\pi, \psi)$ as well. Recall that this solution was obtained by setting all nontrivial inequalities of D to equality.

Now by construction, if we set all the nontrivial inequalities of LP $L(\pi, \psi)$ to equality we get a feasible solution, namely a . Clearly, a and y^* satisfy all complementary slackness conditions.

Therefore they are both optimal. Hence we get:

Lemma 4 For any instance π and monotonically decreasing tradeoff function ψ , y^* is an optimal solution to $D(\pi, \psi)$.

Expanded PaperInterface specification

```
∀ {Primal : Type u_5} {Dual : Type u_6} (primalFeasible : Primal → Prop) (dualFeasible : Dual → Prop)
  (primalObjective : Primal → ℝ) (dualObjective : Dual → ℝ),
  (∀ (primal : Primal) (dual : Dual),
    primalFeasible primal → dualFeasible dual → primalObjective primal ≤ dualObjective dual) →
  ∀ {a : Primal} {ystar : Dual},
    primalFeasible a →
      dualFeasible ystar →
        primalObjective a = dualObjective ystar → ∀ (y : Dual), dualFeasible y → dualObjective ystar ≤ dualObjective y
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma4_dual_optimal_from_primal_dual_match

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 74

Source locator: cited publication, lines 560-577

Verbatim source input

Let $\Delta(\pi, \psi)$ be a $k - 1$ dimensional vector whose i th component is $(\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i)$.
The following lemma relates the right hand side of the LPs L and $L(\pi, \psi)$.
Lemma 5
Proof :
 $l = b + \Delta(\pi, \psi)$.
Consider the i th components of the three vectors. We need to prove:
$$\alpha_1 (1 + \frac{i-1}{k}) + \alpha_2 (1 + \frac{i-1}{k}) + \dots + \alpha_i$$
$$= \frac{iN}{k} + (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i).$$

This equation follows using the fact that $\beta_i = N/k - (\alpha_1 + \dots + \alpha_{i-1})/k$.
[U+0003 control character]

Expanded PaperInterface specification

```
∀ {m : ℕ} (N : ℝ) (alpha beta : Fin m → ℝ) (i : Fin m),
  ∑ j ∈ EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix i, beta j =
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i -
      ∑ j ∈ EconCSLib.Online.MSVV07SourceLemmas.paperRoutePrefix i,
        EconCSLib.Online.MSVV07SourceLemmas.paperRouteDeltaCoeff i j * alpha j →
    EconCSLib.Online.MSVV07SourceLemmas.paperRouteLPRow alpha i =
      EconCSLib.Online.MSVV07SourceLemmas.paperRouteRhs N i +
        EconCSLib.Online.MSVV07SourceLemmas.paperRouteDelta alpha beta i
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma5_tradeoff_rhs_eq_base_add_delta

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 75

Source locator: cited publication, lines 583-593

Verbatim source input

Definition: Let $\text{ALG}(q)$ ($\text{OPT}(q)$) denote the revenue earned by the algorithm (OPT) for query q . Say that a query q is of type i if OPT assigns it to a bidder of type i , and say that q lies in slab i if the algorithm pays for it from slab i .

Lemma 6 For each query q such that $1 \leq \text{type}(q) \leq k - 1$,
 $\text{OPT}(q)\psi(\text{type}(q)) \leq \text{ALG}(q)\psi(\text{slab}(q))$.

Proof : Consider the arrival of q during the run of the algorithm. Since $\text{type}(q) \leq k - 1$, the bidder b to whom OPT assigned this query is still actively bidding from some slab $j \leq \text{type}(q)$ at this time. The inequality in the lemma follows from the criterion used by the algorithm to assign queries, together with the monotonicity of ψ .
[U+0003 control character]

Expanded PaperInterface specification

```
∀ {Slab : Type u_5} [inst : Preorder Slab] (psi : Slab → ℝ) {queryType optCurrentSlab algSlab : Slab}
{optBid algBid : ℝ},
optCurrentSlab ≤ queryType →
  Antitone psi →
    0 ≤ optBid → optBid * psi optCurrentSlab ≤ algBid * psi algSlab → optBid * psi queryType ≤ algBid * psi algSlab
```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma6_per_query_tradeoff

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 76

Source locator: cited publication, lines 597-699

Verbatim source input

Lemma 7

$$\psi(i)(\alpha_i - \beta_i) \leq$$

$$\sum_{i=1}^k$$

Proof :

N

k

We start by observing that for $1 \leq i \leq k - 1$:

$$\text{OPT}(q) = \alpha_i$$

X

$$\text{ALG}(q) = \beta_i$$

$$q:\text{type}(q)=i$$

$$q:\text{slab}(q)=i$$

By Lemma 6

X

$$[\text{OPT}(q)\psi(\text{type}(q)) - \text{ALG}(q)\psi(\text{slab}(q))]$$

$$q:\text{type}(q) \leq k-1$$

$$\leq 0.$$

Next observe that

X

$$\text{OPT}(q)\psi(\text{type}(q))$$

$$q:\text{type}(q) \leq k-1$$

$$=$$

$$\sum_{i=1}^{k-1}$$

X

X

$$\text{OPT}(q)\psi(i)$$

$$\sum_{i=1}^k q:\text{type}(q)=i$$

$$=$$

$$\sum_{i=1}^{k-1}$$

X

$$\psi(i)\alpha_i.$$

$$\sum_{i=1}^k$$

And

X

$$\text{ALG}(q)\psi(\text{slab}(q))$$

$$q:\text{type}(q) \leq k-1$$

$$\leq$$

[form-feed in source extraction]

X

$$\leq$$

$$\text{ALG}(q)\psi(\text{slab}(q))$$

$$q:\text{slab}(q) \leq k$$

X

$$\leq$$

$$\text{ALG}(q)\psi(\text{slab}(q)) +$$

$$q:\text{slab}(q) \leq k-1$$

$$=$$

$$\begin{aligned}
& \sum_{i=1}^{k-1} \text{ALG}(q) \psi(i) + \\
& \sum_{i=1}^{k-1} q \cdot \text{slab}(q) = i \\
& = \\
& \sum_{i=1}^{k-1} \psi(i) \beta_i +
\end{aligned}$$

Expanded PaperInterface specification

```

∀ {m : Type u_5} {Query : Type u_6} [inst : Fintype m] [inst_1 : Fintype Query] (psi alpha beta : m → ℝ)
  (opt alg : Query → ℝ) (queryType querySlab : Query → m) (finalSlabError : ℝ),
  ∑ q, (opt q * psi (queryType q) - alg q * psi (querySlab q)) ≤ 0 →
    ∑ q, opt q * psi (queryType q) = ∑ i, psi i * alpha i →
      ∑ q, alg q * psi (querySlab q) ≤ ∑ i, psi i * beta i + finalSlabError →
        ∑ i, psi i * (alpha i - beta i) ≤ finalSlabError

```

Lean proof endpoint (built separately)

EconCSLib.Online.MSVV07PaperFacing.PaperInterface.section5_lemma7_weighted_perturbation_bound

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation